

Basic Date _____
Changed _____

APOLLO 13	
LM LUNAR SURFACE CHECKLIST	
PART NO	S/N

ME ROOM
BLDG. 45

Basic Date _____

MARCH 23, 1970

APOLLO 13

FINAL

LM LUNAR SURFACE CHECKLIST

MARCH 31, 1970

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This document is under the configuration control of the Crew Procedures Control Board (CPCB). All proposed changes should be submitted to the Apollo Flight Data File Manager, Mr. T. W. Holloway, CF34, Room 230, telephone HU3-4271.

Distribution of this document is controlled by Mr. J. W. O'Neill, Chief, Flight Planning Branch, Flight Crew Support Division.

APOLLO FLIGHT DATA FILE

LM LUNAR SURFACE CHECKLIST

Original 2/16/70

Revision 3/23/70

Change 3/31/70

LIST OF EFFECTIVE PAGES

* INDICATES CURRENT CHANGE

PAGE NUMBER

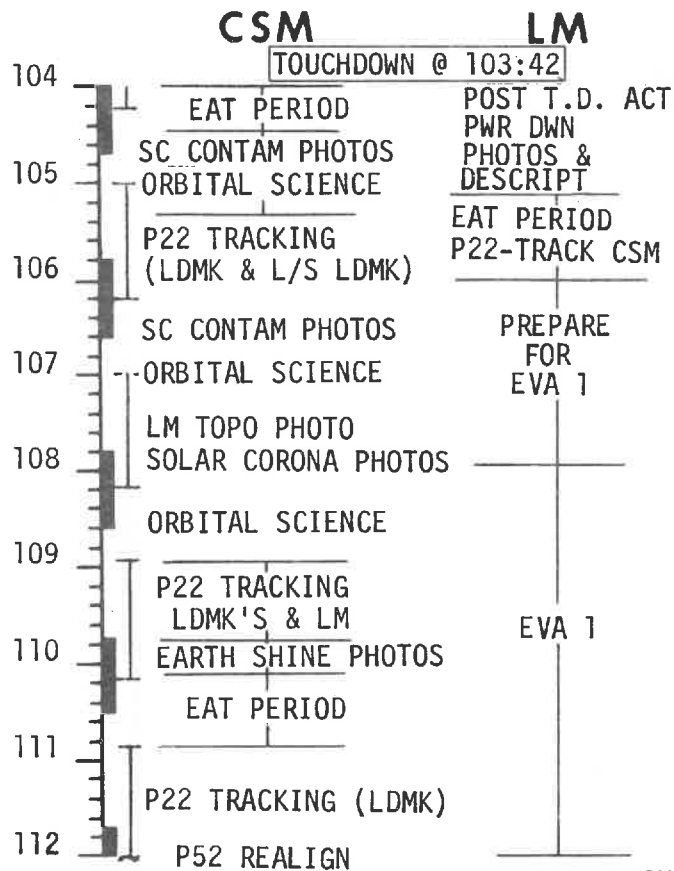
ISSUE

1 thru 13	Basic
*14	3/31/70
15 thru 23	Basic
*24	3/31/70
25 thru 43	Basic
*44	3/31/70
45 & 46	Basic
*47	3/31/70
48	Basic
*49 & 50	3/31/70
51 thru 61	Basic
*62	3/31/70
63 & 64	Basic
*65	3/31/70
66	Basic
*67	3/31/70
68	Basic
*69 & 70	3/31/70
71 & 72	Basic
*73 & 74	3/31/70
75 thru 78	Basic
*79	3/31/70
80 thru 141	Basic
*142	3/31/70
143 thru 152	Basic

23, 39, 55, 81

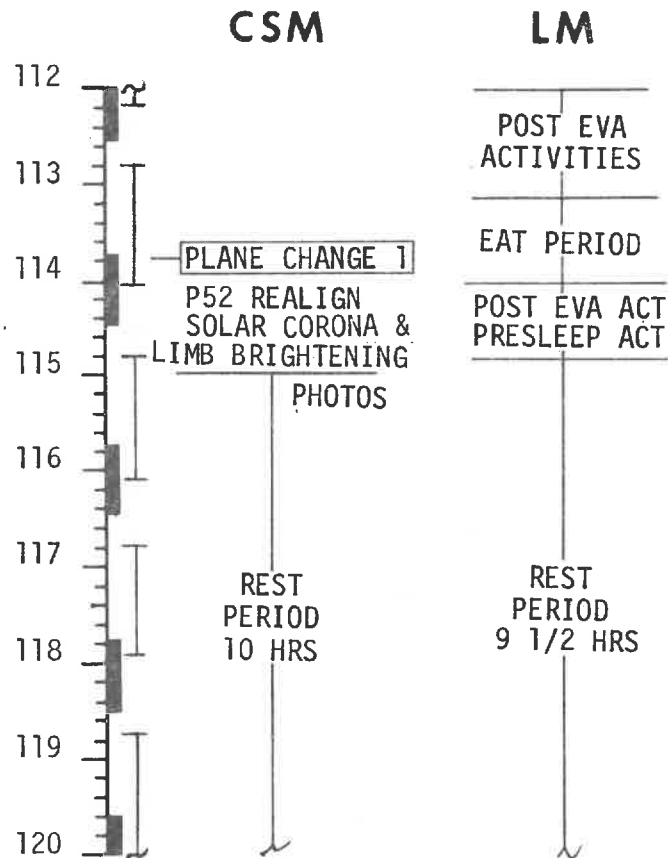
PEN + INK

Basic Date 3/23/70
Changed _____

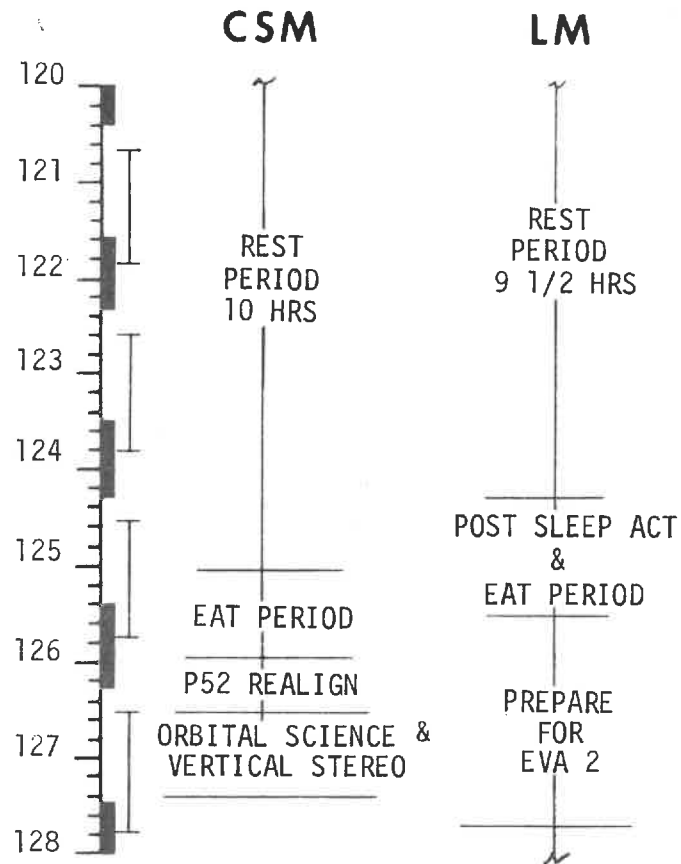


SUR-1

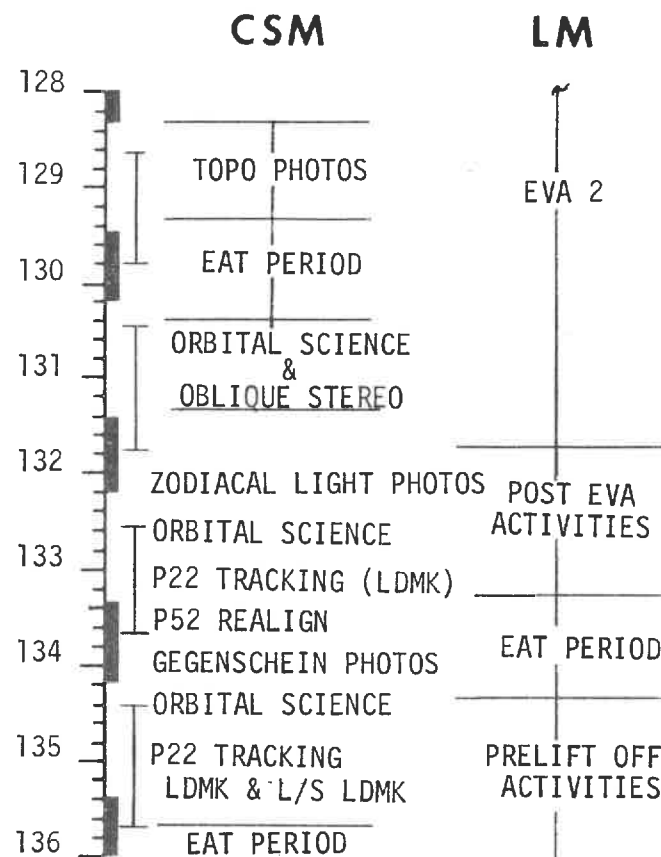
Basic Date _____
Changed _____



Changed _____



SUR-2



LM LIFT OFF @ 137:09

Basic Date 3/23/70

Changed _____

LM-7

Basic Date 3/23/70
Changed _____

FIRST REV ACTIVITY

***** PDI +20 (103:51) *****

CB(11) PGNS: LDG RDR - Open
PRPLNT TEMP/PRESS MON - DES 1,2
Monitor OXID Press Until 20 to
40 psi Then
OXID VENT - CLOSE
MODE CONTROL (Both) - ATT HOLD

Verify INV - 2 Selected
CB(11) INV 1 - Open
DECA PWR - Open

V16 N20E
Record _____ OG
_____ IG
_____ MG

P57, R2 00003
N06 00010
00001
00110
(NO ATT Lt-On/Off, Twice)
N04 + _____ Tilt (.01°)
V32E (Recycle)

CB (16) ASC ECA CONT - Close
BAT 5,6 - OFF/RESET
INVERTER - 2
CB(16) DES ENG OVRD - Open
ASC ECA CONT - Open
CWEA - Open Then Close
(DES. REG-OFF)

047 R _____ Sin Az (To MSFN)
053 R _____ Cos Az (To MSFN)
623 R (+0)

544 R _____ X Gyro Coeff
545 R _____ Y Gyro Coeff
546 R _____ Z Gyro Coeff

400 + 6E Calib Gyros

232 R _____ Ins Alt
465 R _____ Ins HDot
400 R (+0 Calib Complete In
5 min 2 sec)

SUR-3

FIRST REV ACT

N04 _____

PRO

N22 ICDU Angles

PRO (NO ATT LT - On/Off)

N05 _____ Angle Diff (.01°)

PRO

N93 _____ X Torquing Angles (.001°)

Y

Z

V34E, POOE

Verify Cabin Press

PRESS REG A&B - CABIN

CABIN GAS RETURN - AUTO

SUIT GAS DIVERter-Push/CABIN

CABIN REPRESS - AUTO

BIOMED - RIGHT

VHF - OFF, OFF, OFF, OFF

AUDIO (Both): VHF A&B - OFF

S-BD P&Y SET (+112/-43), SLEW,

P _____ Y _____

Doff Helmet & Gloves

Remove & Stow Restraints

Window Shades - Up

CB (2) RR - Close

RR MODE - LGC

V41N72E (00000,28300)

V16N72E

CB (11) RR (2)- Open

V44E

Monitor FUEL Press Until <8 psi Then

FUEL VENT - CLOSE

544 R _____ X Gyro Coeff

545 R _____ Y Gyro Coeff

546 R _____ Z Gyro Coeff

If Gyro Drift Changes >2.0°/hr, AGS Failed

SUR-4

LM-7

Basic Date 3/23/70

Changed _____

LM-7

Basic Date 3/23/70
Changed

***** PDI +:45 (104:16) *****

CB AOT LAMP-Closed

P57, R2 00003

PRO

N06 00010

00002

00110

PRO

1st STAR (ATRIA 134)

Cursor

Spiral

N79 Load Then V32E

Cursor

Spiral

N79 Load Then V32E

Cursor

Spiral

N79 Load Then PRO

SUR-5

Basic Date
Changed

2nd STAR _____ (RASALHAGUE 335)
 Cursor _____
 Spiral _____
 N79 Load Then V32E
 Cursor _____
 Spiral _____
 N79 Load Then V32E
 Cursor _____
 Spiral _____
 N79 Load Then PRO
 N05 _____ Star Angle Diff (.01°)
 PRO _____
 N93 _____ X Torquing Angle (.001°)
 _____ Y
 _____ Z
 PRO (Gyro Torquing)

 N25 00014 ENTR
 N89 _____ Lat (.001°)
 _____ Long/2 (.001°)
 _____ Alt (.01 nm)
 Consult MSFN
 PRO - (UPDATE RLS)
 V34E - (TERM)
 POOE

SUR-6

LM-7

Basic Date 3/23/70
 Changed _____

LM-7

Basic Date 3/23/70
Changed _____

***** PDI +1:00 (104:31) *****

P57, R2 00003

PRO

N06 00010

00002

00110

PRO

1st STAR _____ (ANTARES 233)

Cursor _____

Spiral _____

N79 Load Then V32E

Cursor _____

Spiral _____

N79 Load Then V32E

Cursor _____

Spiral _____

N79 Load Then PRO

SUR-7

Basic Date _____
Changed _____

2nd STAR _____ (FOMALHAUT 645)
Cursor _____
Spiral _____
N79 Load Then V32E
Cursor _____
Spiral _____
N79 Load Then V32E
Cursor _____
Spiral _____
N79 Load Then PRO
N05 _____ Star Angle Diff (.01°)
PRO
N93 _____ X Torquing Angle (.001°)
_____ Y
_____ Z
PRO (Gyro Torquing)

N25 00014 ENTR
N89 _____ Lat (.001°)
_____ Long/2 (.001°)
_____ Alt (.01 nm)
Consult MSFN
PRO - (UPDATE RLS)
V34E - (TERM)
CB AOT - OPEN, CL/0.0°
POOE

SUR-8

LM-7

Basic Date 3/23/70
Changed _____

LM-7

Basic Date 3/23/70
Changed

CB (11) RR (2) - Close
V41N72E (18000,27000)
V16N72E
CB (11) RR (2) - Open
V44E

V40N20E
400 + 3E AGS/PGNS Align
413 + 1E Store Azimuth

047 R Sin Az
053 R Cos Az

STAY/NO STAY From MSFN

CB (16) AEA - Open
AGS STATUS - STBY
CB(16) AEA - Close

***** PDI +1:15 (104:46) *****

UP DATA LINK - DATA
(MSFN Updates RLS & CSM
State Vectors), OFF

Copy Updated P22 Acquisition Time : :
DET - Set Counting Down To Acquisition Time

Window Shades - Down

Configure CB's Per PARTIAL POWER DOWN
Charts SUR-9

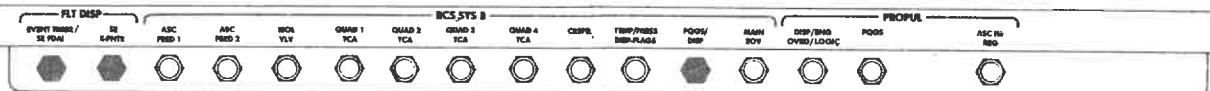
Basic Date
Changed

LM-7

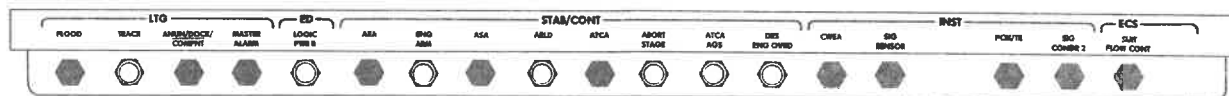
Basic Date 3/23/70
 Changed _____

PARTIAL PWR DN

16



13



7



3



3

SUR-11

Basic Date _____

Changed _____

FDAL 1&2 - INRTL
EARTH/LUNAR - PWR OFF
LTG - OFF
MODE - HOLD/FAST
ALT SET - 45

FUEL & OXID VENT tb-bp
MASTER ARM - OFF
DES VENT - SAFE
ASC He SEL - BOTH
STAGE - SAFE (guarded)

S BAND T/R - T/R
ICS T/R - T/R
RELAY - OFF
MODE - ICS/PTT
AUDIO CONT - NORM
VHF A - OFF
VHF B - OFF
COAS - OFF

TTCA (CDR) - JETS (Dn)

Eng STOP - Reset
Eng START - Reset

TMR CONT - START
OVERRIDE ANUN - OFF
OVERRIDE NUM - OFF
OVERRIDE INTEGRAL - OFF

X POINTER SCALE - HI MULT
RATE/ERR MON - LDG RDR/CMPTR
ATTITUDE MON (CDR) - PGNS
GUID CONT - PGNS
MODE SEL - PGNS
RNG/ALT MON - RNG/RNG RT
SHFT/TRUN - +50°
RATE SCALE - 25°/SEC
ACA PROP (Both) - ENABLE
THR CONT - AUTO
MAN THROT - CDR
ENG ARM - OFF
ATT/TRANSL - 4 JETS
BAL CPL - ON
PRPLNT QTY MON - OFF
PRPLNT TEMP/PRESS MON - ASC
HELIUM MON - PRESS 1
ABORT - Reset
ABORT STAGE - Reset (Guarded)

SUR-12

LM-7

Basic Date 3/23/70
Changed _____

LM-7

Basic Date 3/23/70
Changed _____

TEMP/PRESS MON - OXID MANF
RATE/ERR MON - RNDZ RADAR
ATTITUDE MON (LMP) - AGS
GLYCOL - PUMP 1
SUIT FAN - 1
O2/H2O QTY MON - DES

DES ENG CMD OVRD - OFF
RDR TEST - OFF
TEST MON - AGC
SLEW RATE - HI
RR MODE - LGC
DEAD BAND - MIN

ATTITUDE CONTROL (3) - MODE CONT
MODE CONTROL (Both) - ATT HOLD
IMU CAGE - OFF
EVENT TIMER - DN And START
TEMP MON - RNDZ RDR
RCS SYS A/B-2 QUAD 1,2,3,4 - AUTO
EXTERIOR LTG - OFF
X POINTER SCALE - LO MULT

ACA/4 JET (2) - ENABLE
TTCA/TRANSL (2) - ENABLE

AOT - CL/0.0°
RR GYRO SEL - PRIM

TTCA (LMP) - JETS (Dn)

Eng STOP - Reset
AGS STATUS - STBY

POWER/TEMP MON - CDR BUS
INVERTER - 2
UP LINK SQUELCH - OFF
UP DATA LINK - OFF

AUDIO CONT - NORM
S BAND T/R - T/R
ICS T/R - T/R
RELAY - OFF
MODE - ICS/PTT
VHF A&B - OFF

S BAND MODULATE - PM
XMTR/RCVR - PRIM
PWR AMPL - PRIM
VOICE - VOICE
PCM - PCM
RANGE - RANGE Then CWEA ENABLE

VHF A XMTR & RCVR (2) - OFF
VHF B XMTR & RCVR (2) - OFF
BIOMED - As Desired
TLM - HI
RECORDER - OFF

SUR-13

Basic Date _____

Changed _____

VHF - AFT
TRACK MODE - SLEW
PITCH _____ (From MSFN)
YAW _____ (From MSFN)
S BAND - SLEW

PRESS REG A&B - CABIN
SUIT GAS DIVERter - PUSH/CABIN
CABIN REPRESS - AUTO
PLSS FILL - CLOSE
DES O2 - OPEN
#1, #2 ASC O2 - CLOSE
SUIT ISOL (Both) - SUIT FLOW
SUIT CIRCUIT RELIEF - AUTO
CABIN GAS RETURN - AUTO
CO2 CANISTER SEL - PRIM
PRIM & SEC CO2 CANISTER - CLOSE
WATER SEP SEL - PUSH SEP 1
ASC H2O - CLOSE
SEC EVAP FLOW - CLOSE
PRIM EVAP FLOW No. 2 - CLOSE
DES H2O - OPEN
PRIM EVAP FLOW No. 1 - OPEN
WATER TANK SELECT - DES
SUIT TEMP - As Required
LIQUID COOLING GARMENT - As Required

CABIN RELIEF & DUMP (Both) - AUTO

Unstow Lunar Charts and Maps
Configure (2) 70mm Camrs (RHSSC):
Stow RESEAU Covers In Camr Compt
Stow Polarizing Filter In Camr Comp
One Camr - B&W Mag (ISA)
One Camr - HCEX Mag, 8 /250, 15 ft (ISA)
Stow Dark Slides In Camr Comp
RCU/Camr Adapter Bracket - ISA Top Pocket
Triggers And Handles - RHSSC Camr Pocket
Stow Camr With HCEX Mag In LHSSC
(PLSS LiOH Cartridge Compt)

Describe & Photograph Lunar Surface:
Photo Lunar Surface Out Of Both
Windows Using B&W Film [12]

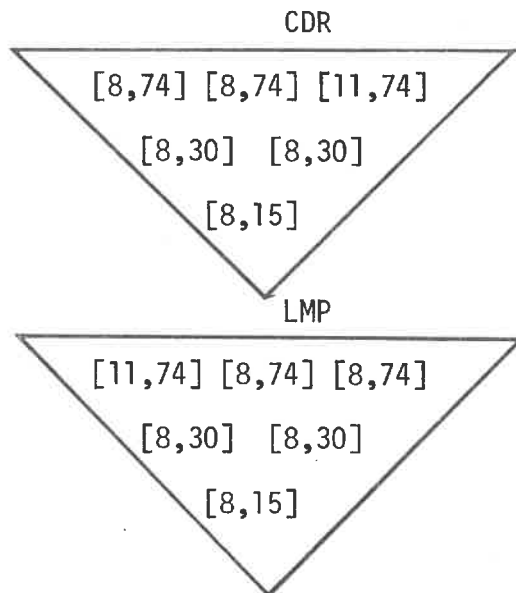
SUR-14

LM-7

Basic Date _____ 3/23/70
Changed _____ 3/31/70

LM-7

Basic Date 3/23/70
Changed _____



Report Features During Descent And
Determine LM Location With HOU (5 Min)
Report Angle Of +Z wrt West. Give Gen-
eral Impression (Earth Analog) And
Predominant Features.

Describe (15 Min)

1 Near Field (define location by
angle and distance from LM)

A Features

- 1 General Surface
- 2 Plains
- 3 Craters
- 4 Rays
- 5 Cones
- 6 Boulder Fields
- 7 Rilles, Faults, Grabens
- 8 Rock Fragments
- 9 Loose Ground-Mass Material
- 10 Coatings

B General Surface

- 1 Texture - smooth, flat,
gentle rolling, rough,
jagged
- 2 Materials - dust, sand,
pebbles, rocks, boulders
[note size, angularity,
and roundness], cinders,
ash fall or flow, lava,
pahoehoe, aa, ejecta

SUR-15

Basic Date _____

- 3 Aerial distribution - uniform, spotted, patterned
- 4 Color/albedo pattern
- 5 Contrasts - abrupt texture or material changes, color/albedo discontinuities, elevation changes [note sharp or diffuse character]
- 6 Origin of surface character - cratering, depositional, flow-like

C Plains

- 1 Extent
- 2 Degree of cratering (age)
- 3 Texture - smooth, flat, gentle rolling
- 4 Color/albedo

D Craters

- 1 Type - rayed (youngest), blocky rim, sharp rim, low rim, subdued, shallow depressions (oldest), chain, dimple
- 2 Size/Shape - diameter, depth (dia/depth ratio), circular, polygonal, square, irregular, elongated
- 3 Ejecta - size, shape, distribution (fields, loops, branches, clusters), material/color/albedo changes, degree of burial
- 4 Color/albedo pattern
- 5 Rim - terraced, hummocky, smooth, radial and concentric patterns, flow patterns, boulder or dune fields, small scale color/albedo variations

SUR-16

LM-7

Basic Date 3/23/70
 Changed _____

LM-7

Basic Date 3/23/70
Changed _____

- 6 Walls - texture, material, small scale color/albedo variations, layers, contacts, strike/dip, bedding, layer thickness and continuity, slump features, flow channels, holes, caves
- 7 Floor - central peak, eruptive features, radial or concentric flow or fracture patterns, rock/boulder fields, small scale color/albedo variations, spatter
- 8 Relation to surrounding craters - chain, cluster, random distribution
- 9 Origin -
Impact: ejecta (direction), central peak, higher rim, rim/wall/floor fragments, impacting material
Volcanic: caldera, flows, cinder spatter
Collapse: no rim or ejecta evidence of material drainage, similar features along linear faults
- E Rays - source, direction, composition, texture/material variations, color/albedo variations, size thickness/width/length ratios
- F Boulder Fields - linear, bunched, sloped, size/angularity/roundness/degree of burial
- G Rilles, Faults, Grabens
- 1 Shape - linear, enechelon, angular, sinuous
- 2 Displacement - relative horizontal and vertical offset of both sides, separation, depth, width
- 3 Age - angularity and slope of sides, fill at bottom, cratering
- 4 Color/Albedo variations

SUR-17

Basic Date _____
Changed _____

5 Walls - texture, material, small scale color/albedo variations, layers, contacts, strike/dip, bedding, layer thickness and continuity, slump features, flow channels, holes/caves

6 Continuity - method of termination, breaks, relative pattern to other similar features, association with other features

H Rock Fragments

1 Size/angularity/roundness

2 Color/albedo relative to surface

3 Height wrt surface - burial, on top, pedestal

4 Surface - visicular, rough, jagged, smooth, layered

5 Distribution - field, cluster, linear group, uniform

I Loose Ground-Mass Material

1 Size - dust, round, gravel, pebbles

2 Sorting - poor, medium, well, bimodal

3 Color/albedo

4 Cohesiveness - loose, friable, cemented, welded

J Coatings

1 Location - windows, LM skin, footpads, rocks, boulders

2 Size - dust, sand, gravel

3 Geometry - uniform, in low spots, rims, fillets, one side only

4 Transport mechanism

SUR-18

LM-7

Basic Date 3/23/70

Changed

LM-7

Basic Date 3/23/70
Changed _____

2 Far Field (define feature location by angle and distance from LM)

A Horizon - flat, smooth, gentle, rolling, scarp (sharp break in slope), jagged, mountains, mesa

B Same as 1B to 1G

3 Ask HOU for questions

Replace B&W Mag In One Camr With HCEX
Settings: LMP 8/250, 5 ft

Stow Camr In LHSSC (PLSS LIOH
Cartridge Compartment)

EAT PERIOD
105:07 to 106:00

LM CONSUMABLES UPDATE

GET (105:00) _____:

RCS A % (77) _____ B (75) _____

02 DES % (91.1) _____ ASC (100) _____

H2O DES % (76.2) _____ ASC (100) _____

A-H DES (1240) _____ ASC (566) _____ SUR-19

105:29 CB(11) RR (2) - Close

V95E

P22E

N06 R2 00001

V83E, Rng <400, PRO, PRO

V16 N72E (18000, 33500)

(N78E, Rng, Rng Rt)

At End Of CSM Track:

N72 Goes To 18000, 27000

P00E

CB(11) RR (2) - Open

V48E

N46 _____ Code

N47 _____ LM wt

V34E, P00E

Notify MSFN of E-Dump

TLM - HI

V74E (42 sec)

P06E

CB(11) IMU OPR - Open

PRO, STBY Lt - On

Remove CB Charts (SUR-20 & 21)

Tape Above CB Panels

Configure CB's Per PWR DN

Basic Date _____
Changed _____

Basic Date 3/23/70
Changed

AC BUS B

AC BUS A

RCS SYS A

FLIGHT DISPLAYS

AC BUS A

PROP

HEATERS

INST

STAB/CONT

ED

LTO

HEATERS RCS SYS A/B-1

ECS

COMM

POINS

EPS

OPEN 1st

★ — OPEN FOR EVA

Basic Date _____
Changed _____

LM-7

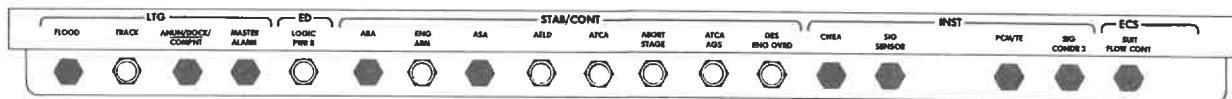
Basic Date 3/23/70
 Changed _____

POWERDOWN & EVA CONFIGURATION

16



14



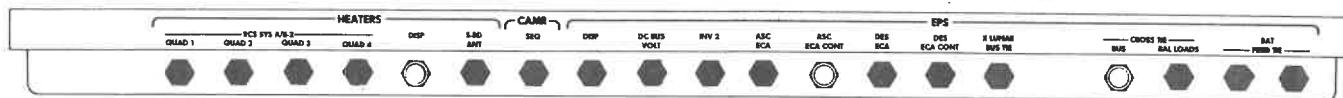
8



3

CLOSE PRIOR TO EGRESS

OPEN PRIOR TO DEPRESS



3

★—OPEN FOR EVA

SUR-21

Basic Date _____

LM-7

Basic Date 3/23/70
Changed

Copy Lift-Off Time in Data
Book For T-n

CREW STATUS REPORT

	CDR	LMP
MED	_____	_____
PRD	_____	_____

CWEA STATUS:

Warning

ASC PRESS

CES AC (Reset via GYRO TEST)

CES DC (Reset via GYRO TEST)

Caution

PREAMPS

SUR-22

EVA 1 PREP

EVA 1 PREP

106:00 CABIN PREP EVA 1

Stow All Loose Items Not Req'd
For EVA

Unstow EVA 1 Prep & Post Card
Remove Transition To One-Man EVA

Page SUR-24, Clip To AOT

Unstow Lunar Surface Map &

Position For EVA

Stow ~~Utility~~ Towel (RHSSC) Above

Data File

106:15 Stow Lunar Surface Checklist

SUR-23

Basic Date 3/23/70

Changed

TRANSITION TO ONE-MAN EVA

BOTH Verify/Perform-As Req'd At Time Of NO GO
PLSS FEEDWATER - CLOSE
Fwd Hatch Closed & Locked (Dump Vlv - AUTO)
CABIN REPRESS - AUTO
CB(16)ECS: CABIN REPRESS - Close(Cab Warn Lt)
Verify Press Increasing
PRESS REGS A & B - CABIN
PLSS O2 - OFF @ Cabin >2.5 Psia
CB(11) ECS: SUIT FAN 1 - Close
CB(16) ECS: SUIT FAN ΔP - Close
ECS Caution & H2O SEP Comp Lts - Out
PGA Press Equal To Cabin, Use Purge Vlv If Req'd
Cab At 5.0 psia, Doff Gloves, Helmets With Visors
PLSS PUMP - OFF
PLSS FAN - OFF

Basic Date 3/23/70
Changed 3/31/70

NO GO CREWMAN:
PLSS MODE - 0
Disconnect - OPS O2 Hose
- Purge Vlv - Stow In Purse
- OPS Actuator From RCU
- RCU From PGA And PLSS
- PLSS COMM, H2O, And O2
Doff PLSS/OPS
Connect LM COMM, O2, & H2O (Audio, Biomed)
Comm Sws - As Req'd

OTHER CREWMAN:
Disconnect PLSS H2O
Connect LM H2O
CB (16) ECS: LCG PUMP - Close

BOTH Stow NO GO Equipment
OPS - Aft Engine Cover (Disconn Antenna)
PLSS - Recharge Station (Remove All 4
Straps, Stow in RHSSC or Exchange On
PLSS's If Required)
RCU - LHSSC

Unstow ONE MAN EVA PREP CARD & LUNAR SURFACE BOOK

SUR-24

Basic Date

LM REPRESS FAILURE PROCEDURE

Verify PRESS REG A&B - EGRESS
 Verify LM Suit Circuit 3.6 - 4.0 Psia
 CB(11) ECS: SUIT FAN 1 - Close
 CB(16) ECS: SUIT FAN ΔP - Close
 ECS Caution & H2O SEP Comp Lts - Out

Verify OPS 02 - OFF
 Disconnect Purge Vlv, Then OPS 02 Hose
 Stow Purge Vlvs In Purse

Connect to LM ECS Hoses, R/R, B/B
 SUIT ISOL - SUIT FLOW
 PLSS FAN - OFF
 PLSS 02 - OFF

Verify Cuff Gage 3.6 - 4.0 Psig
 PGA Diverter Vlvs - Horizontal
 PLSS Mode - 0
 Disconnect PLSS Elec From PGA

Connect To LM Comm (Audio,Biomed)

Audio (CDR & LMP)

VHF A - OFF
 VHF B - OFF
 MODE - ICS/PTT
 RELAY - OFF

COMM:

VHF - OFF, OFF, OFF, OFF, LEFT, HI
 RECORDER - OFF

PLSS Feedwater - Close
 PLSS PUMP - OFF

Disconnect OPS 02 Actuator
 Disconnect RCU From PGA, Then PLSS
 Stow RCU on Mid-Step

Disconnect PLSS H2O From PGA
 Disconnect PLSS Red 02 Hose, Then Blue
 Doff PLSS/OPS, Place on Floor
 Stow OPS 02 Hoses & Actuator

As Req'd-Connect LM H2O to PGA
 CB(16) ECS: LCG PUMP - Close

SUR-25

Basic Date 3/23/70

Changed _____

Basic Date 3/23/70
Changed _____

EQUIPMENT PREP EVA 1

- 1 DET-Set/Down -:45
Unstow PLSS On Floor,
Position Against Hatch
- 1 Stow Cover Bag & Straps In ISA Back Pkt
Stow COAS In FWD Window Mount
Remove Purse From Bottom Pocket ISA,
Position On Panel 5

Empty UCTA's
Check PGA Zippers
PGA Diverter Valves - Vertical

- 1 Fill Drink Bags (Back ISA) & Install

Stow Gas Connector Plugs In Purse
Empty PGA Pockets Into Purse

- 1 Verify Watch On Left Arm, LMP Only
Verify LM O2 Hoses To PGA R/R & B/B

- 1 Unstow LEC And Place On Panel 6
Stow Helmet Bags On Floor

Both Seq Camrs: Settings-2.8/60, ∞ ,12 FR
Verify Operation, Stow Sur Seq Camr In
Recharge Station
Place 2 Seq Camr Mags In Purse

Disconnect Utility Lights & Position On
Aft Eng Cover
Verify Interim Stowage Straps Accessible

Disconnect Seq Camr Bag Lower Hooks
Position Helmets On Aft Eng Cover

CDR Move To Aft Cabin Area
Remove Sur Seq Camr Bag
Unstow Jett Bag, Aft LHSSC
Upper RH Corner

Stow Sur Seq Camr Bag In Jett Bag
Deploy LM EVA Antenna

Unstow B&W TV & Place In Recharge Station
Unstow RCU's (Resnap Flaps) & Place
Inside LHSSC
Stow Helmets On RCU Shelf
Unstow CDR Lunar Boots, Remove &
Stow Purge Valve In Purse
CDR Don Lunar Boots

SUR-26

Basic Date _____
Changed _____

Basic Date _____
Changed _____

LMP Move To Aft Cabin Area
Unstow LMP Lunar Boots, Remove &
Stow Purge Valve In Purse
Stow Utility Lights In Bot Boot Comp
Open Top Boot Comp For EVA Stowage
LMP Don Lunar Boots

CDR Unstow CSRC From LHSSC & Place
In PGA Pocket

Unstow LMP OPS
Remove Pallet, Stow In Jett Bag
Hand LMP OPS To CDR For Checkout
Unstow CDR OPS
Remove Pallet, Stow In Jett Bag

Perform OPS Check (Both)
Stow LMP OPS On Floor
LMP Move To LMP Station
Stow CDR OPS On LH Eng Cover

Position Helmet Bags On Eng Cover
Apply Antifog (CDR Helmet Bag)
CDR Attach Watch To RH EV Glove
Stow Visors, Helmets & EV Gloves On
RH Eng Cover

Position Helmet Bags In SRC Area
Disconnect 3 Armrests, CDR LH &
LMP RH & LH, Place In Jett Bag
Fwd Hatch Handle - UNLOCK

PLSS DONNING -1:07

LMP 1st - Unstow OPS Antenna Lead
& Secure Flap
Attach OPS To Unstowed PLSS
Connect OPS Antenna Lead To PLSS
Verify Sublimator Exhausts Clear

Unstow PLSS Straps & Hoses
Attach Second Yo-Yo (Purse) If Desired |
Remove Dust Cover From PLSS Elect
Conn & Stow In Aft LHSSC
Verify ALL PLSS Valves - Up
Connect Battery Cable

Verify The Following Locked:
OPS To PLSS
OPS Antenna To PLSS
PLSS Battery Connection

SUR-27

Basic Date 3/23/70
Changed _____

Basic Date 3/23/70
Changed

Unstow RCU's From LHSSC & Place
On Eng Cover

Don PLSS/OPS (Lift PLSS Hoses Above
LH Lower Strap)

Connect PLSS O2 Hoses To PGA
Verify DIVERTER, O2, FEEDWATER-OFF

Verify Helmets Accessible

Transfer CDR PLSS To Mid-Step
CDR Repeat PLSS DONNING

Verify RCU Controls:
PUMP, FAN, MODE SEL - OFF
Vol Cont (2) - FULL INCR
(NOTE: Blade-B & AR, Wheel-A)
Connect RCU To PLSS, Then PGA

PLSS COMM CHECK - :44

Verify Powerdown CB Configuration
COMM: MODULATE - FM
CB(16) COMM: TV - Close
Verify Voice Comm With Hou

Audio (CDR)
S-BAND - T/R
ICS - T/R
RELAY - OFF
MODE - VOX (VOX SENS MAX)
VHF A - T/R
VHF B - RCV

Audio (LMP)
S-BAND - T/R
ICS - T/R
RELAY - ON
MODE - VOX (VOX SENS MAX)
VHF A - T/R
VHF B - RCV

COMM:
VHF - VOICE, ON, OFF, ON, OFF, HI
RANGE - OFF/RESET
SQUELCH A&B - Noise Thres + 1-1/2
RECORDER - ON
VHF Antenna - EVA
UPLINK SQUELCH - ENABLE

SUR-28

Basic Date

Changed _____

LMP Connect To PLSS Comm (Audio CB)

PLSS PTT (LMP) - MAIN

PLSS Mode (LMP) - A (Tone-On, Vent

Flag - P, Press Flag - 0)

PLSS 02 Press Gage > 85%

Perform Comm Check With CDR

Note: Unstow PLSS Antenna If It
Transmits Garbled And/Or Loses TM

CDR Connect To PLSS Comm (Audio CB)

Audio (CDR)

VHF A - OFF

VHF B - OFF

No MSFN Reception In PLSS Mode B

PLSS PTT (CDR) - MAIN

PLSS Mode (CDR) - B (Tone-On, Vent

Flag - P, Press Flag - 0)

PLSS 02 Press Gage > 85%

Perform Comm Check With LMP

PLSS Mode (LMP) - B (Tone-On)

PLSS Mode (CDR) - A (Tone-On)

Verify Voice Comm With Each Other

PLSS Mode (Both) - AR (Tone-On)

Perform Comm & TM Check With Hou &

Comm Check With Each Other

Read PLSS 02 Qty to Hou

Note: If Comm Is NO GO With Hou
S-BD MOD - PM
Verify Comm & TM

CB(16) COMM: TV - Open

FINAL SYSTEMS PREP -:34

CB(16) ECS: CABIN REPRESS - Close (Ver)

SUIT FAN Δ P - Open

CB(11) ECS: SUIT FAN 1 - Open

SUIT GAS DIVERTER - PULL-EGRESS

CABIN GAS RETURN - EGRESS

SUIT CIRCUIT RELIEF - AUTO (Verify)

Verify ECS Caution & H2O SEP COMP

Lts - On (~1 Min)

SUR-29

Basic Date 3/23/70

Changed _____

Basic Date 3/23/70
Changed _____

OPS CONNECT -:32

LMP 1st - Unstow OPS 02 Hose
& Actuator

Connect Actuator To RCU

Snap OPS 02 Hose To Side Of PLSS

SUIT ISOL - SUIT DISC

Discon LM 02 Hoses, Secure About PGA

Connect OPS 02 Hose To PGA B/B

Retrieve Purge Valve (Purse)-

Verify Closed & Locked

Install Purge Valve In PGA R/R

Verify PLSS Centered & At

Proper Height

CDR Repeat OPS CONNECT

Drink

DES H2O VLV - CLOSE

HELMET/GLOVE DONNING -:23

Position Mikes (Both)

PLSS FAN - ON (Vent Flag - Clear)

Don Helmets, Then Visors

Check Drink Bag Position

LCG-Cold, As Req'd

CB(16) ECS: LCG PUMP - Open

Disconnect LM H2O Hose

Connect PLSS H2O Hose

Stow LM Hoses (CDR's With Straps To
ECS Module Handhold)

Verify The Following:

Helmet & Visor (2) - Locked &
Adjusted

Torso Tiedown (2) - Adjusted

O2 Connectors (6) - Locked

Purge Valves (2) - Locked

H2O Connectors (2) - Locked

Comm Connectors(2) - Locked

Verify EVA CB Configuration

Verify No Fog RH Window

Tie Jett Bag & Transfer to Eng Cover

Don EV Gloves & Verify:

Wrist Locks (4) - Locked

Glove Straps (4) - Adjusted

If PGA Biting, PLSS O2 - ON/OFF

SUR-30

Basic Date _____

Changed _____

PLSS DIVERTER - MIN (Verify)
PLSS PUMP - ON

PRESS REG A & B - EGRESS

PRESSURE INTEGRITY CHECK -:08

PLSS 02 - ON (Tone-On, 02 Flag-0)
Press Flag Clear (3.1-3.4 Psid)
Cuff Gage 3.7-4.0 Psig
02 Flag Clear

PLSS 02 - OFF (Cuff Gage Decay<.3
Psig In 1 Min)

PLSS 02 - ON (Cuff Gage 3.7-4.0
Psig, Tone & 02 Flag May Come On)

CABIN DEPRESS -:03

Confirm "Go" For Depress From Hou
CB(16)ECS: CABIN REPRESS - Open
CABIN REPRESS VLV - CLOSE

Fwd Dump Valve - OPEN Then AUTO At
3.5 Psia (Verify Cuff Gage Does
Not Drop Below 4.8 Psig)

Verify:

Cabin At 3.5 Psia
LM Suit Circuit 3.6 To 4.3 Psia
PGA > 4.8 Psig & Decaying

Start Wrist Watch :00

Fwd Dump Valve - OPEN

Verify:

Tone-On & H2O Flag-A(1.3-1.6 Psia)
LM Suit Circuit 3.6 To 4.3 Psia
PGA >4.8 Psig & Decaying

Partially Open Fwd Hatch

FINAL PREP FOR EGRESS :03

PLSS FEEDWATER - OPEN (H2O Flag -
Clear In About 4 Min)

Fwd Hatch - Open

SUR-31

Basic Date 3/23/70

Changed _____

Basic Date 3/23/70
Changed

Rest Until Cooling Sufficient

Verify:

PGA 3.7 To 4.8 Psig

LM Suit Circuit 3.6 To 4.3 Psia

CWEA Status:

Warning

ASC PRESS

Caution

PREAMPS

ECS

H2O SEP COMP LT - ON

Lighting: ANUN/NUM - DIM

CB(16) COMM: TV - Close

Release PLSS Antennas

Lower EV Visor :10

SUR-32

EVA 1

CDR CUFF CHECKLIST

PLSS H₂O TO LM H₂O (192)

- LOOSEN PGA TORSO TIEDOWN
- PUMP OFF
- LMP DISCONNECT PLSS H₂O
- ✓ PGA CONNECTOR UNLOCKED
- LMP CONNECT LM H₂O
- CB(16) ECS: LCG PUMP-CLOSED

LM H₂O (192) TO PLSS H₂O

- CB(16) ECS: LCG PUMP-OPEN
- LMP DISCONNECT LM H₂O
- LMP CONNECT PLSS H₂O
- PUMP ON
- TIGHTEN PGA TORSO TIEDOWN

DATE: 3/16/70

CDR - EVA 1CDR EGRESS

0+10 PULL SAFETY & DEPLOY MESA [JETTISON BAG
TO CDR

TOSS BAG TO -Y AREA
DROP LEC TO GND.

ASCENT ✓

0+18 FAM

CAMERA TRANSFER [TRANSFER

LM ✓ & REPORT

CONTINGENCY SAMPLE

STOW INTACT ON STRUT

PHOTO AREA

0+25 LMP EGRESS AID [EGRESS

TEMP STOW 70 MM [FAM

CAM OR GIVE

TO LMP

DATE: 3/12/70

SUR-33

Basic Date 3/23/70

Changed _____

Basic Date 3/23/70
Changed _____

0+30	<u>S-BAND DEPLOY (+Z)</u> ORIENT TOWARDS EARTH SAVE THERMAL COVER STEADY LEG FOR DISH DEPLOY & DUCK CONNECT CABLE CALL LMP TO ASSIST <u>ANTENNA ALIGN</u> WATCH PLSS ANT/DISH	[FAM [TV	DATE: 3/12/70
0+45	<u>FINISH ETB PREP</u> DEPLOY TABLE UNSTOW ETB - TAKE OUT BAGS & SAFETY - STOW ON MESA LOAD IN ETB - LiOH CANS PLSS BATTS CONT SAMPLE BAG HOOK UP LEC	[INGRESS	

DATE: 3/12/70		<u>S-BAND ALIGN</u> TRANSFER ETB TO LMP HANG ETB ON SIDE MESA PUT ON MOVIE CAMERA PHOTOS LMP EGRESS MOVIES OR STILLS GIVE MOVIE CAMERA TO LMP <u>FLAG DEPLOY</u> UNSTOW FROM MESA UNSTOW HAMMER DRIVE MAST EXTEND HORIZ. ROD PLACE IN MAST PUT ON 70 MM CAMERA (ON FOOTPAD) CAP TV LENS & PICK UP TV	[ANT. SW. [MAG FOR EXPEND [EGRESS [TV REORIENT [FILM CDR
	1+00		

SUR-34

Basic Date _____

1+10	WALK TO SEQ BAY VIA +Y - LM INSPECT TV REPOSITION TO SEQ BAY (f:44 DIST:25 ANGLE:25) & UNCAP KEEP STRUTS OUT OF PICTURE <u>ALSEP PREP</u> OPEN DOOR (WHITE) PKG 1 OUT HANDLE UP TOOLS OFF PKG 2 & IN HTC DRILL OFF PLACE ON PAD <u>RTG FUELUP</u> PUTTERS IN PKGS PKG 2 DOWN TO FUEL MAST TOGETHER - ✓ LOCK MAKE DUMBBELL	[TO SEQ BAY DATE: 3/12/70			
	<table border="1"> <tr> <td>TO DEPLOY ALSEP</td> </tr> <tr> <td>BOOM: Pull White Till Pkg Clear; Ratchet Stripes To Gnd; Pull Pin Hockey Stick</td> </tr> <tr> <td>MANUAL: Pull Release; Pull String; Haul Out Pkg; Pull Pin Hockey Stick</td> </tr> </table>	TO DEPLOY ALSEP	BOOM: Pull White Till Pkg Clear; Ratchet Stripes To Gnd; Pull Pin Hockey Stick	MANUAL: Pull Release; Pull String; Haul Out Pkg; Pull Pin Hockey Stick	
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BOOM: Pull White Till Pkg Clear; Ratchet Stripes To Gnd; Pull Pin Hockey Stick					
MANUAL: Pull Release; Pull String; Haul Out Pkg; Pull Pin Hockey Stick					

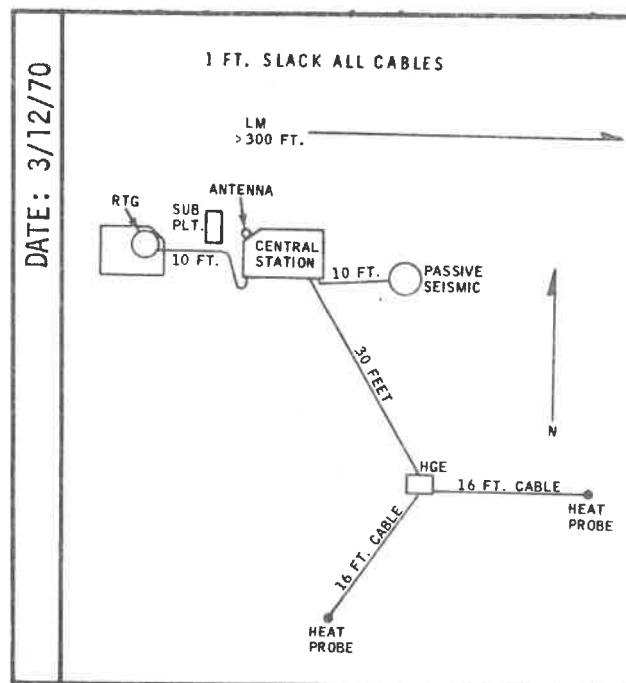
DATE: 3/12/70	BOOMS IN SHUT DOORS (STRIPES) 1+25 TV PAN - AT f:44 UNLESS GND ADVISES PHOTO PANS GET DUMBBELL - MEET LMP PUT ON TOTE BAGS 1+30 TRAVERSE > 300 FT WEST
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SUR-35

Basic Date 3/23/70
Changed _____

Basic Date 3/23/70
 Changed _____

1+35 ALSEP DEPLOY DISCON MAST FROM PKG 2 } PUTTERS RELOC PKG 2 10 FT. WEST } NORTH 70 MM CAM TO LMP EMPLACE PKG 2 REPORT TEMPS ON ALL EQUIP. DEPLOY HORSE COLLAR & CONNECT CABLE READ AMPS <u>DO NOT PUSH SHORT BUTTON</u> EMPLACE PKG 1 & EYEBALL ALIGN GET TONGS FROM LMP 1+45 AID HFE REMOVAL [HFE REMOVAL LEAN TONGS & MAST ON PKG 1 DEPLOY PSE STOOL (DIG HOLE) REMOVE SUB PALLET, STOW MAST DEPLOY PSE (PULL OFF BOLTS) LEVEL, REPORT DEGREES	DATE: 3/12/70
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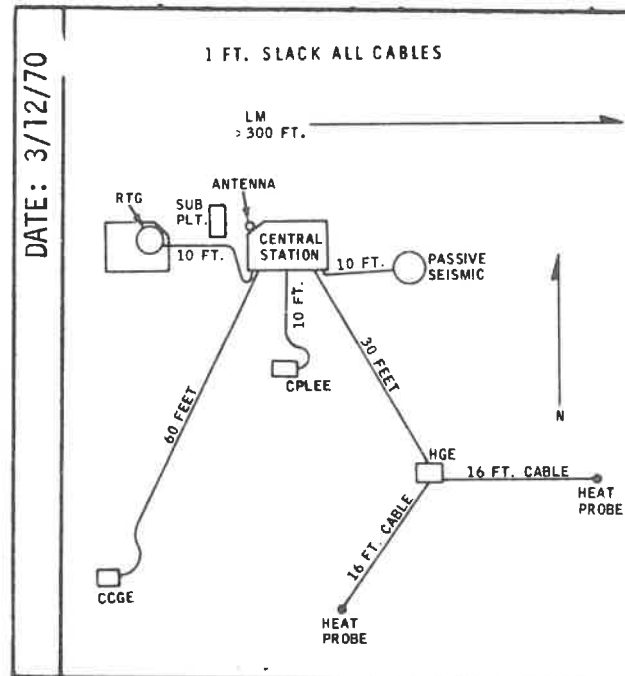


SUR-36

Basic Date _____
 Changed _____

Changed _____

REMOVE CCGE & PULL PIN ON REEL. DEPLOY 60 FT. SW. PUTTER IS GNOM DEPLOY CPLE (PULL PIN) TAMP DIRT, PUTTER IS GNOM 2+10 GNOM & BUBBLE ALIGN PKG 1 FREE ANT. CABLE RELEASE ALL SUNSHIELD BOLTS (INNER LAST) ✓ CORNERS & DEPLOY SUNSHIELD (WATCH ANT. AIM BOX) MAST IN SOCKET & ASSEMBLE ANT. ALIGN ANT: LEVEL, SHADOW, AZ <u>16.13</u> ELEV <u>6.40</u> ✓ ALIGN PICK UP BOLTS & WEIGHT PSE SKIRT (IF REQUIRED)	DATE: 3/12/70
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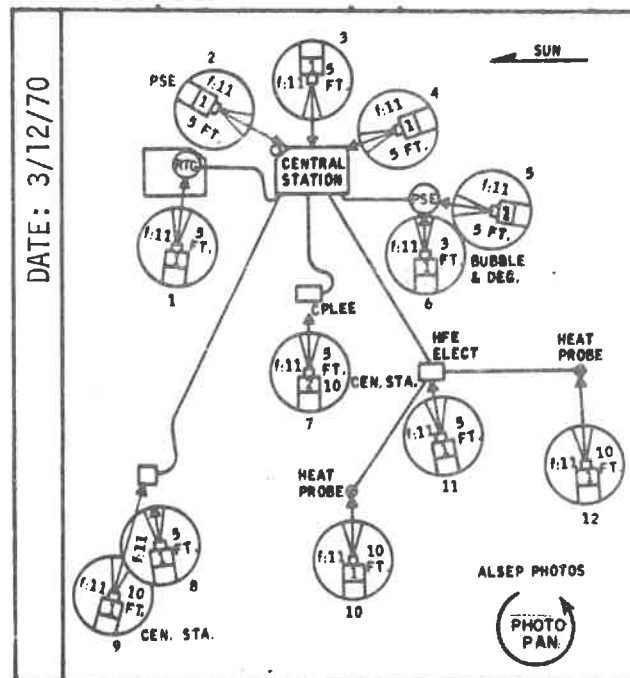


SUR-37

Basic Date 3/23/70
 Changed _____

Basic Date 3/23/70
 Changed _____

(IF HFE NOT DEPLOYED) <u>DEPLOY HFE</u> REPORT RAMMER MARKS (INNER/OUTER) PROBE & HOLE SHIELD EMPLACE HOLE COVER 2+35 <u>ACTIVATE CEN STA</u> [HFE DEPLOYED] (WATCH AMPS) PRESS BUTTON TURN SW 1 (GND MARK) <u>PHOTO LAYOUT</u> → [CORE DRILL ALL: 1/250TH GET MOVIE CAMERA FILM LMP DRILLING <u>CORE SAMPLE REMOVAL</u> [CORE REMOVAL] REPORT CAPS & STEM ORDER START AT TOP & WORK DOWN TO BIT WIPE STEMS ALL DRILL DEBRIS > 8 FT SE STRIP GLOVES 3+05 EMU ✓	DATE: 3/12/70
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SUR-38

Basic Date _____

Basic Date _____
Changed _____

3+10	RETURN TRAVERSE COLLECT SELECTED SAMPLES - PHOTO STEREO OR CROSS SUN BEFORE & 1 SHOT AFTER ROCKS - 75% FINES - 25% (OR 10 LBS.) 3+20 CAP LENS & REPOINT TV AT MESA OR UNCAP TV LENS END TRAV. SRC 2 & CSC OFFLOAD LOAD IN ETB: MAP CAMERAS LENS OFFLOAD SAMPLES FROM TOTE BAGS INTO WEIGH BAGS STOW WEIGH BAG IN SRC 1 MESA WEIGH BAG ON SCALE FILL BAG WITH SAMPLES, SOIL, ROCKS & STOW (USE SIEVE) STOW CORE STEMS REMOVE SRC SEAL	DATE: 3/12/70
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3+35	CLOSE SRC 1 RETURN ITEMS & SAMPLES IN CDR TOTE BAG HANG LMP TOTE BAG ON MESA 3+40 CLEAN EMU'S LEC TRANSFERS TRANSFER ETB TRANSFER CDR TOTE BAG TRANSFER SRC 1 TONGS IN HTC HTC IN SUN 3+52 EVA TERMINATION -CAP TY CLEAN EMU ATTACH LEC TO PORCH INGRESS RAISE EV VISOR 4+00 REPRESS	DATE: 3/12/70 [INGRESS [ETB, TOTE [BAG & SRC [STOW [LEC TO CDR
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SUR-39

Basic Date 3/23/70
Changed _____

Basic Date 3/23/70
Changed _____

LMP CUFF CHECKLIST

<u>PLSS H₂O TO LM H₂O (192)</u> ◦ LOOSEN PGA TORSO TIEDOWN ◦ PUMP OFF ◦ CDR DISCONNECT PLSS H₂O ◦ ✓ PGA CONNECTOR UNLOCKED ◦ CDR CONNECT LM H₂O ◦ CB (16) ECS: LCG PUMP-<u>CLOSED</u> <u>LM H₂O (192) TO PLSS H₂O</u> ◦ CB(16) ECS: LCG PUMP-<u>OPEN</u> ◦ CDR DISCONNECT LM H₂O ◦ CDR CONNECT PLSS H₂O ◦ PUMP ON ◦ TIGHTEN PGA TORSO TIEDOWN	DATE: 3/16/70
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	<u>LMP - EVA 1</u> 0+10 PASS JETTISON BAG EGRESS TO CDR PASS LEC TO CDR 0+18 PHOTO FAM (FAM CAMERA TRANSFER PHOTO CONTING SAMP. SHADE f2.8 @ 1/60, 12 fr SUN f8 @ 1/60, 12 fr 0+25 <u>EGRESS</u> 70 MM CAMERA IN DOORWAY CLOSE HATCH ASCENT ✓ <u>FAM</u> - TAKE 70 MM CAMERA FROM CDR - PLACE ON FOOT PAD
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SUR-40

Basic Date _____

Basic Date _____
Changed _____

0+32	<u>TV DEPLOY</u> [S-BAND RELEASE MESA SNAPS PEEL MESA UNSNAP ADAPTER TRIPOD DEPLOY PULL 2 SNAPS TV BOX STRAP RELEASE 2 LENS STRAPS SET LENS f:44 & <u>CAP LENS</u> , ALSC SW-PEAK TV ON TRIPOD UNSTOW CABLE POSITION 50 FT. OFF +Y SET LENS - f:44; DIST:50; ANGLE:25 UNCAP LENS <u>CAUTION</u> KEEP LENS $\geq 45^\circ$ TO SUN	DATE: 3/12/70
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DATE: 3/12/70	<u>START ETB PREP</u> [S-BAND <u>DEPLOY TABLE</u> UNSTOW ETB - TAKE OUT BAGS & SAFETY - STOW ON MESA LOAD IN ETB - ◦ LIOH CANS ◦ PLSS BATTS ◦ CONT. SAMPLE BAG HOOK UP LEC <u>ASSIST CDR ANT. ALIGN</u> 0+45 <u>INGRESS</u> [S-BAND ✓ TV CKT BKR ✓ S-BAND - LUNAR STAY TRACK MODE - OFF [ETB PREP <u>TRANSFER ETB (UP)</u> [TRANSFER
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SUR-41

Basic Date _____
Changed _____ 3/23/70

Basic Date 3/23/70
Changed _____

<u>OFFLOAD CONSUMABLES</u> STOW BATTS (OPS AREA) STOW CANS (ASC. ENG. COVER) STOW CONT. SAMP (BOTTOM BOOT) <u>IN ETB</u> - SEQ. CAM, 2 MAGS, TV CAM, AND MAP <u>TRANSFER ETB (DOWN)</u>	DATE: 3/12/70
1+00 <u>EGRESS</u> CAMERA IN DOORWAY CLOSE HATCH DESCEND TAKE MOVIE CAMERA <u>FLAG DEPLOY</u> [UNSTOW TV REORIENT FILM CDR - SEQ CAM	

DATE: 3/12/70	1+10 <u>WALK TO SEQ BAY</u> [TO SEQ BAY VIA -Y - LM INSPECT. <u>ALSEP PREP</u> OPEN DOOR (WHITE) PKG 2 OUT HANDLE UP PULL 4 PINS HTC DEPLOY TOOLS IN HTC DRILL OFF PLACE ON PAD SEQ CAM IN HTC <u>RTG FUELUP</u> TILT CASK & REMOVE DOME [PKG 2 DOWN CAPSULE INTO RTG TAKE HTC & DRILL TO MESA		
	<table border="1"><tr><td>TO DEPLOY ALSEP</td></tr><tr><td>BOOM: Pull White Till Pkg Clear; Ratchet Stripes To Gnd; Pull Pin Hockey Stick</td></tr><tr><td>MANUAL: Pull Release; Pull String; Haul Out Pkg; Pull Pin Hockey Stick</td></tr></table>	TO DEPLOY ALSEP	BOOM: Pull White Till Pkg Clear; Ratchet Stripes To Gnd; Pull Pin Hockey Stick
TO DEPLOY ALSEP			
BOOM: Pull White Till Pkg Clear; Ratchet Stripes To Gnd; Pull Pin Hockey Stick			
MANUAL: Pull Release; Pull String; Haul Out Pkg; Pull Pin Hockey Stick			

SUR-42

Basic Date _____

1+25 <u>SOLAR WIND IN LIOH HOLE</u> <u>CSC IN SUN</u> <u>MESA TO HTC</u> [TV, PHOTO PANS MAP, TONGS, GNOMON EXT. HANDLE, HAMMER WRENCH SAFETY LINE IN LMP TOTE BAG UNLOAD SRC 1 SCALE & BAG ON MESA IN HTC - CORE STEMS BAG DISPENSER CORE STEM CAPS SEAL ORGANIC SAMPLE, PADS TO SIDE B & W TV CAM IN SRC HOLE ✓ HTC CONTENTS DEPLOY SOLAR WIND IF FEASIBLE (+Y 60 FT.) REORIENT TV PUT ON TOTE BAG 2 16 MM MAGS CDR TOTE BAG PICK UP HTC & DRILL 1+30 <u>TRAVERSE</u> >300 FT. WEST TAKE MOVIES OF CDR	DATE: 3/12/70
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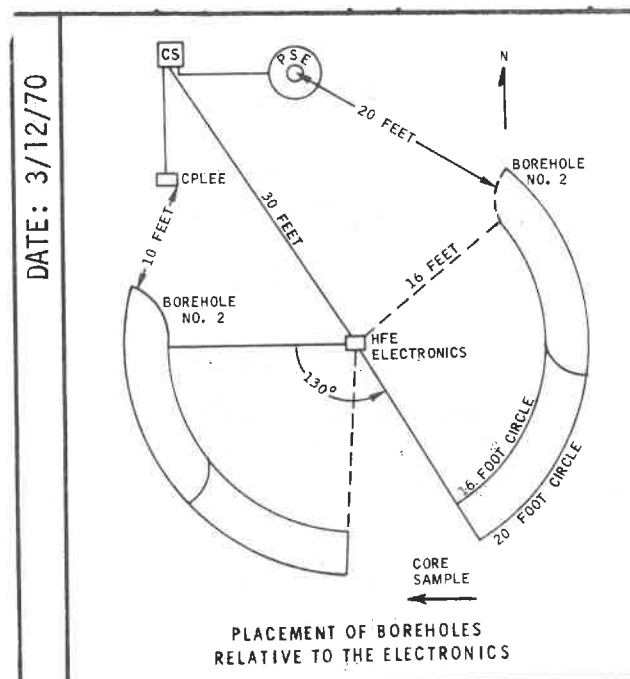
1+35 <u>ALSEP DEPLOY</u> <u>DEPOSIT HTC & DRILL</u> PKG 1 & 2 <u>VICINITY DRILL SITE</u> POSITION 16 MM CAM IN HTC PLACE DRILL ON HTC TAKE TONGS TO CDR REMOVE CDR 70 MM CAMERA HELP CDR WITH HOOKUP TAKE MAST OFF PKG 1 & STOW 1+45 REMOVE HFE RETURN TO HTC & DRILL EYEBALL ALIGN ELECT. E-W DETACH PROBE BOX & SPLIT LEAN 1/2 BOX ON HFE ELECT.	DATE: 3/12/70
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SUR-43

Basic Date _____
 Changed _____ 3/23/70

Basic Date 3/23/70
 Changed 3/31/70

DEPLOY HFE CABLES & LOCATE HOLES (LEAVE PROBES IN BOXES) 1+55 <u>DRILL ASSEMBLY</u> <u>BORE HOLE DRILLING</u> TO RELEASE STEM - 1/4 TURN CCW CW BACK, BLIP DOWN, BLIP UP <u>HFE DEPLOY</u> REPORT RAMMER MARKS (INNER/OUTER) PROBE, HOLE SHIELD, TOP SHIELD VERIFY ELECT ALIGN [PHOTO ALSEP <u>CORE SAMPLE</u> CCW TO LOCK TREADLE. BREAK JOINTS- WRENCH CCW ABOVE JOINT, CW BELOW, THEN RAISE STRING. CAP TOP & BIT - WIPE STEMS REPORT CAPS & STEM ORDER START AT TOP & WORK DOWN TO BIT ADVISE HOUS - THROW DRILL ALL DRILL DEBRIS > 8 FT. SE STRIP GLOVES RAMMER DOWN HOLE	DATE: 3/12/70
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SUR-44

Basic Date _____

3+05 EMU ✓ 3+10 <u>RETURN TRAVERSE</u> TAKE MOVIE CAMERA COLLECT SELECTED SAMPLES PHOTO DOWN SUN BEFORE LANDMARK SHOT X SUN - (BEFORE OR AFTER) ROCKS - 75 % FINES - 25 % (OR 10 LBS.)	DATE: 3/12/70	3+20 <u>SRC 1 PACK</u> OR END TRAV. PLACE HTC AT MESA OFFLOAD SAMPLES FROM TOTE BAGS PLACE RETURN ITEMS CDR TOTE CLOSE CDR TOTE - PLACE ON FOOTPAD ✓ MAGS IN POUCH REMOVE SRC 2 - PLACE ON +Y FOOTPAD SHADY SIDE, LID DOWN SUN PLACE S-BAND ANT COVER SRC 2 3+30 <u>LOAD ETB OR CHECK:</u> MAP HAND LENS CAMERAS <u>DEPLOY SOLAR WIND (+Y 60 FT.)</u> (UNLESS DONE)	DATE: 3/12/70
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SUR-45

Basic Date 3/23/70
Changed

Basic Date 3/23/70
Changed _____

<u>EVA TERMINATION</u>	
CLEAN EMU	[TRANSFER PREP
TONGS IN HTC	
3+40	INGRESS
	RAISE EV VISOR
	LEC PREP
	<u>LEC TRANSFERS</u>
	STOW ETB & TOTE BAG
	STOW SRC 1
	[ASCENT
	HAND OUT LEC TO CDR
3+52	[INGRESS
4+00	REPRESS

SUR-46

EVA 1 POST

POST EVA 1

Verify Bag Jettisoned

PLSS FEEDWATER - CLOSE

Fwd Hatch - Close & Lock

Dump Valves (Both) - AUTO

Note: PLSS O2 & PRESS Flags May Come On During Repress. If PLSS O2 <10% Manually Control Cabin Repress To Maintain Positive PGA Pressure. (Leave Cabin Repress CB Open For Manual Repress)

Lighting: ANUN/NUM - BRIGHT

CABIN REPRESS - AUTO

CB(16)ECS: CABIN REPRESS - Close
(MASTER ALARM & Cabin Warning
Lt - On)Verify Press Increasing
PRESS REG A & B - CABIN

PLSS O2 - OFF @ Cabin > 2.5 Psia
CABIN REPRESS Valve Closes At 4.4 Psia
Verify Cabin Press Stable At 4.6
To 5.0 Psia (Cabin Warning Lt-Off)
Operate OPS Purge Valve To Depress
Suit As Required
DET - Set/up :00

POST EVA SYSTEMS CONFIGURATION :00

CABIN GAS RETURN - AUTO
SUIT CIRCUIT RELIEF - AUTO (Verify)
SUIT GAS DIVERter - PUSH-CABIN

Verify EVA CB Configuration
CB(11) ECS: SUIT FAN 1 - Close
CB(16) ECS: SUIT FAN ΔP- Close
(ECS Caution & H2O SEP Comp Lts - Out)
CB(16) COMM: TV - Open

Doff Gloves, Helmets With Visors &
Stow On Eng Cover

DES H2O VLV - OPEN
Remove Purge Valve & OPS O2 Hose
Stow Purge Valves In Purse

SUR-47

Basic Date 3/23/70Changed 3/31/70

Basic Date 3/23/70
Changed _____

Connect LM 02 Hoses

SUIT ISOL (Both) - SUIT FLOW
PLSS PUMP - OFF
PLSS FAN - OFF

Disconnect PLSS H2O From PGA
Connect LM H2O To PGA
CB(16) ECS: LCG PUMP - Close
(Note: Adjust LCG Cooling Gradually)

PLSS Mode (Both) - 0
Connect To LM Comm (Audio, Biomed)

AUDIO (CDR & LMP)

VHF A - OFF
VHF B - RCV
MODE - ICS/PTT
RELAY - OFF

COMM:

VHF - OFF, OFF, OFF, ON, LEFT, HI
RECORDER - OFF
UPLINK SQUELCH-OFF

PLSS 02 RECHARGE :10

Verify DES 02 >35%

Connect 02 Supply To PLSS (LMP's 1st)
PLSS FILL - OPEN Then CLOSE After 2 Min

PLSS Mode - AR (02 QTY ~75%)
PLSS Mode - 0

Repeat 02 Recharge For CDR PLSS

Stow 02 Supply Hose

PLSS/OPS DOFFING :16

Disconnect OPS Actuator From RCU
Disconnect RCU From PGA
Verify All RCU Controls - Off
Disconnect RCU From PLSS & Stow On
Mid-Step

Disconnect PLSS 02 Hoses
Doff PLSS/OPS

SUR-48

Basic Date

Stow LMP PLSS On Floor
 Stow CDR PLSS On Mid-Step
 Stow OPS 02 Hose, Actuator, & Antenna
 Disconnect OPS Antenna Connector

Install Gas Connector Plugs (Purse)
 In PGA

Caution: Replace Expended PLSS LiOH
 Carts & Batts Numbered 1 or 2 with
 New Carts & Batts Numbered 3 or 4

Replace CDR PLSS Batt, Stow In Fwd LHSSC
 Connect Cable to Battery
 Stow PLSS Hoses
 Change LiOH Cart, Temp <130°, Read Decal
 Remove OPS & Stow Antenna Connector
 Stow OPS On Eng Cover
 Stow CDR PLSS In Recharge Station

Stow One RCU Inside LHSSC, One Outside
 Stow EV Gloves Above Comm Panels
 Stow Helmets & Visors In Helmet Bags
 Stow Helmet Bags On Floor

Replace LMP PLSS Batt, Stow In Fwd LHSSC
 Connect Cable to Battery
 Stow PLSS Hoses
 Change LiOH Cart, Temp <130°, Read Decal
 Remove OPS & Stow Antenna Connector
 Stow PLSS On Floor

Verify OPS Bottle Press (Both)
 Place LMP OPS On Floor

POST EVA CABIN CONFIGURATION :54

CDR Move To Aft Cabin
 Report SRC ____ & Tote Bag #1 Wt ____ To
 Houston (Scale Bottom ISA)
 Stow SRC In Lower Comp
 Stow CDR OPS In Top Comp
 Snap Tote Bag
 Stow Tote Bag In Cover Bag
 (Back ISA) If Req'd
 Stow Tote Bag On Aft Eng Cover

Configure Sur Seq Camr (Mag, Settings
 8/250,∞,6FR)
 Remove Mags From ETB & Stow In RHSSC
 Remove Map, Hand Lens, Stow As Required

SUR-49

Basic Date ____ 3/23/70
 Changed ____ 3/31/70

Basic Date 3/23/70
Changed 3/31/70

Replace 70mm Camr Mags With B&W
Stow Return Items In ISA
Stow In ETB:

70mm Camrs (2)
Sur Seq Camr (1)
Sur Seq Camr Mags (2)
B&W Mag (1)
Polarizing Filter
Map (1)

Unstow Jett Bag From Aft LHSSC,
Upper RH Corner
Place ETB In Jett Bag, Stow On RH
Cabin Floor, Fwd

Verify Powerdown CB Configuration
MODULATE - PM
CB(16) ECS: LCG PUMP - Open
Reverse O2 Hoses, R/B & B/R

Unstow Lunar Surface Checklist SUR-50
Stow EVA 1 Prep & Post Card

1:09

EAT PERIOD
113:09 TO 114:00

Copy Liftoff Time In Data Book
For T - n

PLSS RECHARGE 2:00

Perform Feedwater Collection:
Unstow Feedwater Collection Bags
Aft LHSSC, Upper LH Corner
Remove Spring Scale From Bag

Flatten Bag To Remove Trapped Gas
Zero Spring Scale
Weigh RCU, Report To Hou
Stow RCU Inside LHSSC
Connect Bag To PLSS H2O Fill

PLSS O2 - ON
After 30 sec, PLSS H2O VLV - OPEN
Drain Feedwater Bladder 1.5 Minutes
PLSS H2O VLV - CLOSE
PLSS O2 - OFF

SUR-50

Basic Date _____

Changed _____

Disconnect Bag From PLSS H2O Fill
Weigh Bag & Record LBS, CDR _____
LMP _____ (Report To Hou)
Stow Bag In LHSSC, Fwd Section
Repeat For 2nd PLSS
Stow Scale In LHSSC, Fwd Section

Verify 1 Hr Elapsed Since Initial
O2 Recharge (DES O2 >35%)

Connect O2 Supply To PLSS
PLSS FILL - OPEN Then CLOSE
After 10 min

Perform Feedwater Recharge (Decal)

Verify PLSS FILL - CLOSED
Disconnect O2 Supply

Repeat O2 & H2O Recharge For Second PLSS

DES H2O Vlv - OPEN
DET-Stop 2:30

EVA DEBRIEFING WITH HOU (5 MIN)
Report Status Of PLSS Recharge

CREW STATUS REPORT		
	CDR	LMP
MED	_____	_____
PRD	_____	_____

VOICE - DN VOICE BU
PWR AMPL - OFF
VHF ANTENNA - AFT

SUR-51

Basic Date 3/23/70
Changed _____

Basic Date 3/23/70
Changed _____

Configure Sleep Stations and Stowage:

Disconnect LMP Restraint Cables From
Stowage

CDR Unstow CDR Hammock & Move To
AFT Cabin, Route Hoses Behind PGA
Doff Lunar Boots

LMP Stow Equipment On Floor Per Diagram
Place Cue Card Over AOT For Sleep
Unstow Towels (Hammock Pouch)

LMP Configure 1-6 & Ingress Hammock

Notes: 1 & 2 - Adjust For Height

5 - Route Under Hoses & Through
Vertical Handhold On Aft
ECS Panel

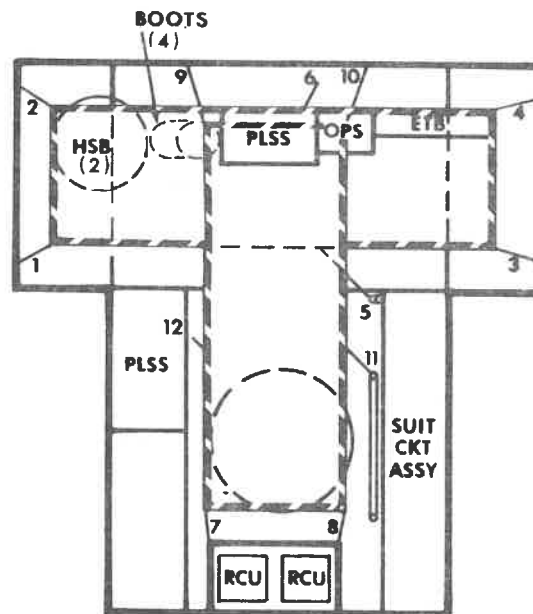
6 - To Lower RH ISA Attach Point

CDR Configure 7-11, Ingress Hammock,
Configure 12 & Adjust

Notes: 9 & 10 - Route Under CDR Hoses,
Adjust For Height

11 - Through Horizontal Handhold
on AFT ECS Panel

12 - Upper Outboard Donning
Station Pin



SUR-52

Basic Date _____
Changed _____

Basic Date _____
Changed _____

REST PERIOD
114:50 TO 124:20

CREW STATUS REPORT
CDR _____
LMP _____

Crew Awake - Verify CWEA Status:
Warning: ASC PRESS, Caution: PREAMPS
PWR AMPL - PRIM
VOICE - VOICE
Unstow Jett Bag, Aft LHSSC,
Upper RH Corner
Stow Hammocks In Jett Bag
Stow LMP Restraint Cables
CB(16) ECS: LCG PUMP - Close

MED _____
PRD _____

Copy Liftoff Time In Data
Book For T - n

Change LM ECS LiOH Cartridge

Stow Used LiOH Cart & Bracket In Jett Bag
Stow Helmet Bags On Eng Cover

PRO, V37E06E, PRO
(STBY Lt - ON)

EAT PERIOD
124:30 TO 125:30

LM CONSUMABLES UPDATE

GET (124:30) _____:

RCS A % (77) _____ B (75) _____

O2 DES % (68.1) _____ ASC (100) _____

H2O DES % (45.1) _____ ACS (100) _____

A-H DES (798) _____ ASC (566) _____

STAY/NO STAY For EVA 2 Prep

PRE EVA 2 PLANNING WITH HOU (15 MIN)

SUR-53

Basic Date 3/23/70
Changed _____

Basic Date 3/23/70
Changed

125:45 CABIN PREP EVA 2

Stow All Loose Items Not Req'd For EVA
Unstow EVA 2 Prep & Post Card

126:15 Stow Lunar Surface Checklist

SUR-54

EVA 2 PREP

EQUIPMENT PREP EVA 2

DET-Set/Down -:30

Empty UCTA's

Check PGA Zippers

PGA Diverter Valves - Vertical

Fill Drink Bags & Install

Stow Gas Connector Plugs In Purse

Empty PGA Pockets Into Purse

Verify Watch On Left Arm

Verify LM O2 Hoses To PGA R/R & B/B

Don Lunar Boots (CDR 1st)

Transfer Rt Angle Bkt & Remote Cntl

Cable To RHSSC

Unsnap Fwd LHSSC For Jett - Leave 1 Snap

Stow Utility Lights If Required

Stow Helmet Bags On LH Cabin Floor

Verify 70mm Camrs Configured In ETB,

Settings CDR 8/250, LMP 11/250

Stow Jett Bag In LHSSC

Verify Sur Seq Camr - Configured

Settings 8/250,∞, 12FR

CDR Move To Aft Cabin

Unstow CDR OPS

Perform OPS Check (Both)

Stow Both OPS On Floor

FWD Hatch Handle - UNLOCK

CDR Move To CDR Station

Position Helmet Bags On Eng Cover

Apply Antifog (CDR Helmet Bag)

Stow Visors & Helmets On RH Eng Cover

Position Helmet Bags In SRC Area

PLSS DONNING -1:07

LMP 1st - Unstow OPS Antenna Lead

& Secure Flap

Attach OPS To Unstowed PLSS

Connect OPS Antenna Lead To PLSS

Verify Sublimator Exhausts Clear

SUR-55

Basic Date 3/23/70Changed 3/31/70

Basic Date 3/23/70
Changed _____

Unstow PLSS Straps & Hoses
| Attach Second Yo-Yo (Purse) If Desired
Verify ALL PLSS Valves - Up

Verify The Following Locked:
OPS To PLSS
OPS Antenna To PLSS
PLSS Battery Connection

Unstow RCU's From LHSSC & Place
On Eng Cover

Don PLSS/OPS (Lift PLSS Hoses Above
LH Lower Strap)

Connect PLSS 02 Hoses To PGA
Verify DIVERTER, 02, FEEDWATER-OFF

Verify Helmets Accesible

Transfer CDR PLSS To Mid-Step
CDR Repeat PLSS DONNING

Verify RCU Controls:
PUMP, FAN, MODE SEL - OFF
Vol Cont (2) - FULL INCR
(NOTE: Blade-B & AR, Wheel-A)
Connect RCU To PLSS, Then PGA

PLSS COMM CHECK --:44

Verify Powerdown CB Configuration
COMM: MODULATE - FM
CB(16) COMM: TV - Close
Verify Voice Comm With Hou

Audio (CDR)
S-BAND - T/R
ICS - T/R
RELAY - OFF
MODE - VOX (VOX SENS MAX)
VHF A - T/R
VHF B - RCV

Audio (LMP)
S-BAND - T/R
ICS - T/R
RELAY - ON
MODE - VOX (VOX SENS MAX)
VHF A - T/R
VHF B - RCV

SUR-56

Basic Date _____

Changed _____

COMM:

VHF - VOICE, ON, OFF, ON, OFF, HI

RANGE - OFF/RESET

SQUELCH A&B - Noise Thres + 1-1/2

RECORDER - ON

VHF Antenna - EVA

UPLINK SQUELCH - ENABLE

LMP Connect To PLSS Comm (Audio CB)

PLSS PTT (LMP) - MAIN

PLSS Mode (LMP) - A (Tone-On, Vent

Flag - P, Press Flag - 0)

PLSS 02 Press Gage >75%

Perform Comm Check With CDR

Note: Unstow PLSS Antenna If It
Transmits Garbled And/Or Loses TM

CDR Connect To PLSS Comm (Audio CB)

Audio (CDR)

VHF A - OFF

VHF B - OFF

No MSFN Reception In PLSS Mode B

PLSS PTT(CDR) - MAIN

PLSS Mode (CDR) - B (Tone-On, Vent

Flag - P, Press Flag - 0)

PLSS 02 Press Gage >75%

Perform Comm Check With LMP

PLSS Mode (LMP) - B (Tone-On)

PLSS Mode (CDR) - A (Tone-On)

Verify Voice Comm With Each Other

PLSS Mode (Both) - AR (Tone-On)

Perform Comm & TM Check With Hou &

Comm Check With Each Other

Read PLSS 02 Qty to Hou

Note: If Comm Is NO GO With Hou
S-BD MOD - PM
Verify COMM & TM

CB(16) COMM: TV - Open

SUR-57

Basic Date 3/23/70

Changed _____

Basic Date 3/23/70
Changed _____

FINAL SYSTEMS PREP - :34

CB(16) ECS: CABIN REPRESS - Close (Ver)
SUIT FAN ΔP - Open
CB(11) ECS: SUIT FAN 1 - Open
Verify ECS Caution & H2O SEP COMP
Lts - On (~1 Min)

SUIT GAS DIVERTER - PULL-EGRESS
CABIN GAS RETURN - EGRESS
SUIT CIRCUIT RELIEF - AUTO (Verify)

OPS CONNECT - :32

LMP 1st - Unstow OPS 02 Hose & Actuator
Connect Actuator To RCU
Snap OPS 02 Hose To Side of PLSS
SUIT ISOL - SUIT DISC
Discon LM 02 Hoses, Secure About PGA

Connect OPS 02 Hose To PGA B/B
Retrieve Purge Valve Purse
Verify Closed & Locked
Install Purge Valve In PGA R/R
Verify PLSS Centered & At Proper Height

CDR Repeat OPS CONNECT

Verify Items Prepared For Jettison:
ECS LiOH Cartridge-Jett Bag
Hammocks-Jett Bag
LHSSC (Fwd Section)-PLSS Batteries,
LiOH Carts, Food Waste, Urine Bags
Position ETB On Mid-Section Step

Drink
DES H2O VLV - CLOSE

HELMET/GLOVE DONNING - :23

Position Mikes (Both)
PLSS FAN - ON (Vent Flag - Clear)
Don Helmets, Then Visors
Check Drink Bag Position

LCG - COLD, As Req'd
CB(16) ECS: LCG PUMP - Open
Disconnect LM H2O Hose
Connect PLSS H2O Hose
Stow LM Hoses

SUR-58

Basic Date _____

Basic Date _____
Changed _____

Verify The Following:

Helmet & Visor (2) - Locked &
Adjusted
Torso Tiedown (2) - Adjusted
O2 Connectors (6) - Locked
Purge Valves (2) - Locked
H2O Connectors (2) - Locked
Comm Connectors (2) - Locked

Verify EVA CB Configuration

Verify No Fog RH Window

Tie Jett Bag & Transfer To Eng Cover

Don EV Gloves & Verify:

Wrist Locks (4) - Locked

Glove Straps (4) - Adjusted

If PGA Biting, PLSS O2 - ON/OFF

PLSS DIVERter - MIN (Verify)

PLSS PUMP - ON

PRESS REG A & B - EGRESS

PRESSURE INTEGRITY CHECK -:08

PLSS O2 - ON (Tone-On, O2 Flag-0)

Press Flag Clear (3.1-3.4 Psid)

Cuff Gage 3.7-4.0 Psig

O2 Flag Clear

PLSS O2 - OFF (Cuff Gage Decay <.3
Psig In 1 Min)

PLSS O2 - ON (Cuff Gage 3.7-4.0
Psig, Tone & O2 Flag May Come On)

CABIN DEPRESS -:03

Confirm "Go" For Depress From MSFN

CM(16)ECS: CABIN REPRESS - Open

CABIN REPRESS VLV - CLOSE

Fwd Dump Valve - OPEN Then AUTO At
3.5 Psia (Verify Cuff Gage Does
Not Drop Below 4.8 Psig)

Verify:

Cabin At 3.5 Psia

LM Suit Circuit 3.6 To 4.3 Psia

PGA >4.8 Psig & Decaying

SUR-59

Basic Date 3/23/70
Changed _____

Basic Date 3/23/70
Changed

Start Wrist Watch :00

Fwd Dump Valve - OPEN

Verify:

Tone-On & H2O Flag - A (1.3-1.6 Psia)

LM Suit Circuit 3.6 To 4.3 Psia

PGA >4.8 Psig & Decaying

Partially Open Fwd Hatch

FINAL PREP FOR EGRESS :03

PLSS FEEDWATER - OPEN (H2O Flag -
Clear In About 4 Min)

Fwd Hatch - Open

Rest Until Cooling Sufficient

Verify:

PGA 3.7 To 4.8 Psig

LM Suit Circuit 3.6 To 4.3 Psia

CWEA Status:

Warning

ASC PRESS

Caution

PREAMPS

ECS

H2O SEP COMP LT - ON

Lighting: ANUN/NUM - DIM

CB(16) COMM: TV - Close

Jettison Bag & LHSSC

Release PLSS Antennas

Lower EV Visor :10

SUR-60

EVA 2

CDR CUFF
CHECKLISTDISTANCE ESTIMATION

IF LM TOP TO PAD:

WIDE AS EARTH - 620 FT.

ECLIPSED BY THUMB - 350 FT.
(ARM'S LENGTH)

IF LM CLUSTER TO CLUSTER:

WIDE AS EARTH - 350 FT.

ECLIPSED BY THUMB - 200 FT.
(ARM'S LENGTH)

IF ASCENT STAGE - TOP TO INTERFACE:

WIDE AS EARTH - 280 FT.

ECLIPSED BY THUMB - 160 FT.
(ARM'S LENGTH)

DATE: 3/12/70

SUR-61

Basic Date 3/23/70Changed

Basic Date 3/23/70
Changed 3/31/70

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DATE: 3/12/70	<u>CDR - EVA 2</u> 0+10 <u>CDR EGRESS</u> JETTISON BAGS HAND LEC TO LMP DESCEND ETB TRANSFER DOWN PUT 70 MM CAMERA ON RCU 0+20 GET SRC 2 [LMP EGRESS PLACE ON TABLE & OPEN BRING HAND TOOL CARRIER TO MESA
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SUR-62

Basic Date

0+25	<u>SRC OFFLOAD</u> GAS, SPEC ENVIRON, MAG SAMPLES IN LMP TOTE BAG POUCH CORE CAPS IN CLIP HTC 35 BAG DISPENSER IN HTC CORE TUBES IN HTC RACK CLOSE ORG. SAMPLE - BACK IN SRC <u>ITEMS FOR HTC</u> FOXHOLE SHOVEL, HAMMER, GNOM, MAP <u>PUT TOTE BAG ON LMP</u> SAFETY LINE & FILTER IN POUCH TONGS ON YOYO 70 MM CAMERA ON RCU ASSEMBLE SMALL SCOOP & EXT HANDLE GET CSC & HAVE MESA BRUSH NEAR <u>THERMAL DEGRADATION SAMPLE</u> SAMPLE 1: CSC 1 SIDE, THEN DIRT ON SAMPLE. CSC BOTH SIDES. BRUSH OFF SAMPLE, CSC BOTH SIDES	DATE: 3/17/70
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DATE: 3/17/70	(HOLDS SAMPLES SAMPLE 2: DIRT ON SAMPLE, CSC BOTH SIDES SMALL SCOOP EXT HANDLE IN HTC <u>SIEVE SAMPLE 5 MINUTES</u> SELECT & PHOTO TYPICAL AREA COLLECT INTO EXTRA WEIGH BAG SEAL BAG & STOW IN ETB <u>CONTAMINATED SAMPLE (UNDER QUAD 3)</u> TAKE X SUN STEREO BEFORE COLLECT FINES & <u>DOUBLE BAG</u> TAKE X SUN PHOTO & STOW IN ETB <u>COLOR CHART (LMP HOLDS)</u> PLACE GNOMON NEAR ROCK POSITION CHART DOWN SUN 5 FT. TAKE 4 PHOTOS: f:5.6, 8, 11, 16 TAKE 1 PHOTO f:11 45° AZIMUTH HOLD CHART FOR LMP TAKE CSC CAMERA & START TRAVERSE
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SUR-63

 Basic Date 3/23/70
 Changed _____

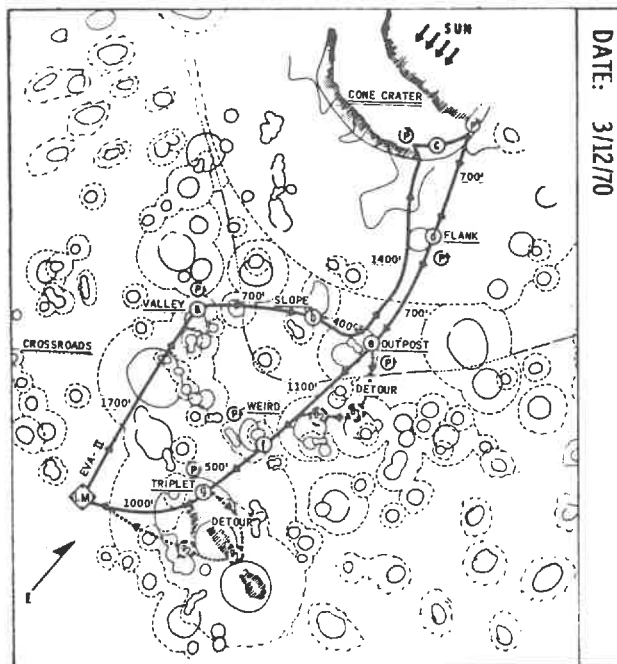
Basic Date 3/23/70
Changed _____

<u>TRAVERSE</u> <u>REPORT:</u> ALL MOVEMENTS INCLUDING DIRECTIONS BETWEEN SAMPLES LOCATION WRT LM PHOTOS OTHER THAN NOMINAL SAMPLE BAG NUMBERS OBSTRUCTIONS WRT LM- GET BEHIND & TRY COMM (CDR) LMP PHOTO OBSTRUCTION, CDR, LM <u>PHOTOS</u> PAN AT EACH LEG, SITE, DIRECTION CHANGE--MAP SHOWS NOMINAL CSC CAMERA PICTURES-PIGGY BACK ON DOC. SAMP. AS REQUIRED <u>REPORT ORIENT & FRAME NO.</u> <u>CRATER DOCUMENTATION</u> SERIES OF SAMPLES RADIALY FROM CENTER	DATE: 3/12/70
--	----------------------

DATE: 3/12/70	CORE SAMPLES (SINGLE & DOUBLE) TAKE ON SURFACE ONLY(SEE MAP) BEFORE: REPORT S/N TUBE(S) DRIVE TUBE(S) & PHOTO X SUN 15 FT AFTER: PHOTO X SUN STEREO 5 FT <u>TRENCHES</u> PHOTO SITE X SUN, DOWN SUN DIG DOWN SUN OFF 10° f:5.6,1/125 DOCUMENTED SAMPLES BOTTOM-TOP FILL, BOTTOM, TOP, SIDES,DISCONS CSC CAMERA PHOTOS BOTTOM RESET 70 MM CAMERA 1/250 <u>COLLECT 1 LARGE ROCK- FOOTBALL</u> USE COLOR CHART <u>COLLECT FILLET MAT'L & ROCK</u> DOCUMENT & USE CSC CAMERA <u>SAMPLE ROCK TRAIL</u> FINES IN & BESIDE-CHIP FROM ROCK USE CSC CAMERA SAMPLE CRATER RIM CRESTS
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SUR-64

Basic Date _____



DATE: 3/12/70

DATE: 3/24/70

VALLEY: DOC. SAMP, PAN [7 MIN.
SURF CHARACT.

SLOPE: DOC. SAMP, [10 MIN.
CORE PATTERNED GND
COMPARE WITH VALLEY

CONE: 2 PANS RIM 300 FT. BASE
POLARIZED PIX [30 MIN.
ROLL BOULDER -24 fps MOVIE
BOULDER TRACKS & RADIAL
VARIATIONS, TRY COMM.

FLANK: DOC. SAMP & PAN [7 MIN.
COMPARE ROCKS W/CONE

OUTPOST: DO CUFF CK LIST [30 MIN.

WEIRD: DOUBLE CORE MULT. EJECTA
DOC. SAMP SUPERIMPOSED
ORIGIN ELONG. CRATER [15 MIN.

TRIPLET: DOC. SAMP, PAN [7 MIN.
PATTERNED GND

SUR-65

Basic Date _____ 3/23/70
Changed _____ 3/31/70

Basic Date 3/23/70
Changed _____

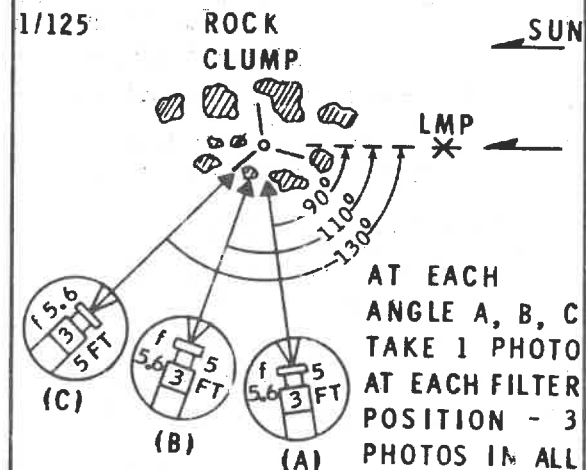
POLARIMETRIC AREA SAMPLE

1. LOCATE CLUMP OF ROCKS
2. GNOMON IN CLUMP
3. TELL LMP PHOTO ROCKS DOWN SUN BEFORE & AFTER SAMPLING
4. ATTACH FILTER & RESET CAMERA f5.6, 1/125
5. GO TO CROSS SUN 5 FT (A)
6. PERFORM STEPS A, B, C
7. COLLECT DIFFERENT KINDS OF ROCKS IN CLUMP (AT LEAST 4)
8. TAKE FAR SHOTS & DISCARD FILTER
9. RESET CAMERA 1/250

DATE: 3/12/70

POLARIMETRIC AREA SAMPLE

SHUTTER
1/125



REPORT FILTER POSITION EACH SHOT

3/12/70

SUR-66

Basic Date _____

OUTPOST		DATE: 3/25/70	DATE: 3/12/70	COMMENT ON SOIL PHYSICAL, MECH PROPS
1. COLLECT GAS ANALYSIS SAMPLE (SMALL CAN): SEVERAL CLEAN ROCKS, GLASSES, SPLATTERED ROCKS, VARIETY DESIRED, DOCUMENT COLLECTION AS USUAL				TAKE DOC. SAMPLES BOTTOM TO TOP, SIDES, DISCONS, FILL. CSC CAMERA IN BOTTOM, SIDES DOCUMENT BOOT PRINT IN FILL
2. COLLECT MAGNETIC SAMPLE (TEFLON CAN) 2-3 ROCKS MICROBRECCIA & CRYSTALLINE DOCUMENT COLLECT AS USUAL				4. COLLECT SPEC. ENVIRON SAMPLE (LARGE CAN): FINE-BOTTOM DEEP TRENCH - DOCUMENT COLLECTION
3. DIG DEEP TRENCH (SOIL MECH.) DOCUMENT BEFORE. THEN f:5.6, 1/125 DIG 2 FT. DEEP 10° OFF DWN SUN DOCUMENT BEFORE SAMPLING: LMP STAND DWN SUN EDGE TRENCH CDR STEREO PR CROSS SUN BOTH SIDES, THEN 1 SHOT DWN SUN LMP 1 SHOT UP SUN				5. (LMP) a. USE CSC CAMERA b. RADIAL FINE SAMPLES SW OUTPOST CRATER - 3 @ 10 FT. INTERVALS - PHOTO AFTER
				6. (CDR) GO BEHIND ROCK OR INTO CRATER TO TEST COMM.
				7. USE COLOR CHART WITH LARGE ROCKS, COLOR CONTRASTS
				8. COLLECT FOOTBALL

SUR-67

Basic Date 3/23/70
 Changed 3/31/70

Basic Date 3/23/70
Changed _____

CSC CAMERA TIPS

REPORT SUN COMPASS/MAINTAIN STD
ORIENTATION. REPORT FRAME NO.
TAKE:
UNEXPECTED FEATURES (TAKE SAMPLE)
GLASSY FEATURES (TAKE SAMPLE)
ROCK/SOIL JUNCTION-UP/DOWN HILL
(TAKE SAMPLE ROCK & FILLETS)
UNDISTURBED SURFACES (LEVEL & SLOPE)
SURFACE PATTERNS -STRIATIONS
ROCK SURFACES-FISSURES,CLEAVAGES,
BEDDING, CRYSTALLIZATION, COLOR
CONTRASTS
SMALL CRATERS ON ROCKS & SURFACES
"LEE" OF ROCKS (FROM ROCKET BLAST)

DATE: 3/12/70

3+20 EVA CLOSEOUT

PLACE WEIGH BAG ON SCALE
TAKE OFF LMP TOTE BAG
BAGGED SAMPLES IN WEIGH BAG
SEAL WEIGH BAG 1 [TAKE OFF
& STOW CDR TOTE
WEIGH BAG 2 ON BAG
SCALE
BAGGED & UNBAGGED SAMPLES IN
WEIGH BAG
SEAL WEIGH BAG 2 & STOW
STOW REMAINING SAMPLES IN
TOTE BAG
SPECIAL, GAS SAMPLES
IN SRC [SOLAR WIND
ROLLED
MAG SAMPLE IN TOTE BAG

DATE: 3/12/70

SUR-68

Basic Date _____

Changed _____

LMP CUFF
CHECKLIST

		DATE: 3/25/70	DISTANCE ESTIMATION
	MAP IN ETB		IF LM TOP TO PAD:
	CORE TUBES IN SRC		WIDE AS EARTH - 620 FT.
	REMOVE SRC SEAL	[COLLECT	ECLIPSED BY THUMB - 350 FT.
	CLOSE & SEAL SRC	EXTRA BAG	(ARM'S LENGTH)
	COLLECT EXTRA BAG 50 FT. FROM LM		
	CAMERAS (2) IN ETB		IF LM CLUSTER TO CLUSTER:
	EXTRA MAG IN ETB		WIDE AS EARTH - 350 FT.
	SEQ CAMERA MAGS (3) IN ETB		ECLIPSED BY THUMB - 200 FT.
3+40	CSC CAM MAG IN ETB	[INGRESS	(ARM'S LENGTH)
	✓ETB LIST (FLAP) &		
	PREP FOR TRANSFER		IF ASCENT STAGE - TOP TO INTERFACE:
	TRANSFER ETB & TOTE	[ETB &	WIDE AS EARTH - 280 FT.
	TRANSFER SRC 2	SRC STOW	ECLIPSED BY THUMB - 160 FT.
	EVA TERMINATION		(ARM'S LENGTH)
	CLEAN EMU - TAKE OFF TONGS		
	JETTISON LEC		
3+52	INGRESS	[LEC TO CDR	
4+00	REPRESS		

SUR-69

Basic Date 3/23/70
Changed 3/31/70

Basic Date 3/23/70
Changed 3/31/70

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DATE: 3/12/70	LMP - EVA 2 <u>ETB CONTENTS</u> 2 CAMERAS, 3 MAGS (B&W) 1 MOVIE CAMERA, 3 MAGS POLARIZING FILTER MAP JETTISON BAGS TO CDR HOOK UP LEC 0+15 <u>ETB TRANSFER DOWN</u> 0+20 <u>EGRESS</u> ◦ CLOSE HATCH ◦ DESCEND ◦ PUT 70 MM CAMERA ON RCU
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SUR-70

Basic Date _____

0+25	GET CSC CAMERA DEPLOY SKIRT TAKE LM VICINITY CSC PHOTOS	DATE: 3/17/70
0+27	PUT TOTE BAG ON CDR RIG LENS BRUSH ON HTC TONGS ON YO YO SAFETY & FILTER IN LMP TOTE	
0+30	MOVIE CAMERA IN HTC <u>THERMAL DEGRADATION SAMPLE</u> SAMPLE 1: BAG IN CDR TOTE UNBAG & UNFOLD - HOLD OUT SHAKE WHEN DIRTY - HOLD OUT HOLD WHEN BRUSHED & CSC [CSC REFOLD & REBAG & BRUSH	
	SAMPLE 2: UNBAG & UNFOLD SHAKE WHEN DIRTY - HOLD OUT REFOLD & REBAG [CSC BAG IN ETB	

DATE: 3/17/70	SPARE MAGS IN CDR TOTE <u>SIEVE SAMPLE 5 MINUTES</u> SELECT & PHOTO TYPICAL AREA COLLECT INTO EXTRA WEIGH BAG SEAL BAG & STOW IN ETB <u>CONTAMINATED SAMPLE (QUAD 3)</u> TAKE DOWN SUN BEFORE HOLD BAG FOR CDR & <u>DOUBLE BAG</u> STOW IN ETB (CDR MAY DO) <u>COLOR CHART</u> GET FROM HTC & HOLD FOR CDR GIVE TO CDR & CHANGE PLACES POSITION CHART DOWN SUN 5 FT. TAKE 4 PHOTOS: f:5.6, 8, 11, 16 TAKE 1 PHOTO f:11 45° AZIMUTH GET CHART & STOW IN HTC <u>PICK UP HTC & START TRAVERSE</u>
---------------	--

SUR-71

Basic Date 3/23/70
Changed _____

Basic Date 3/23/70
Changed _____

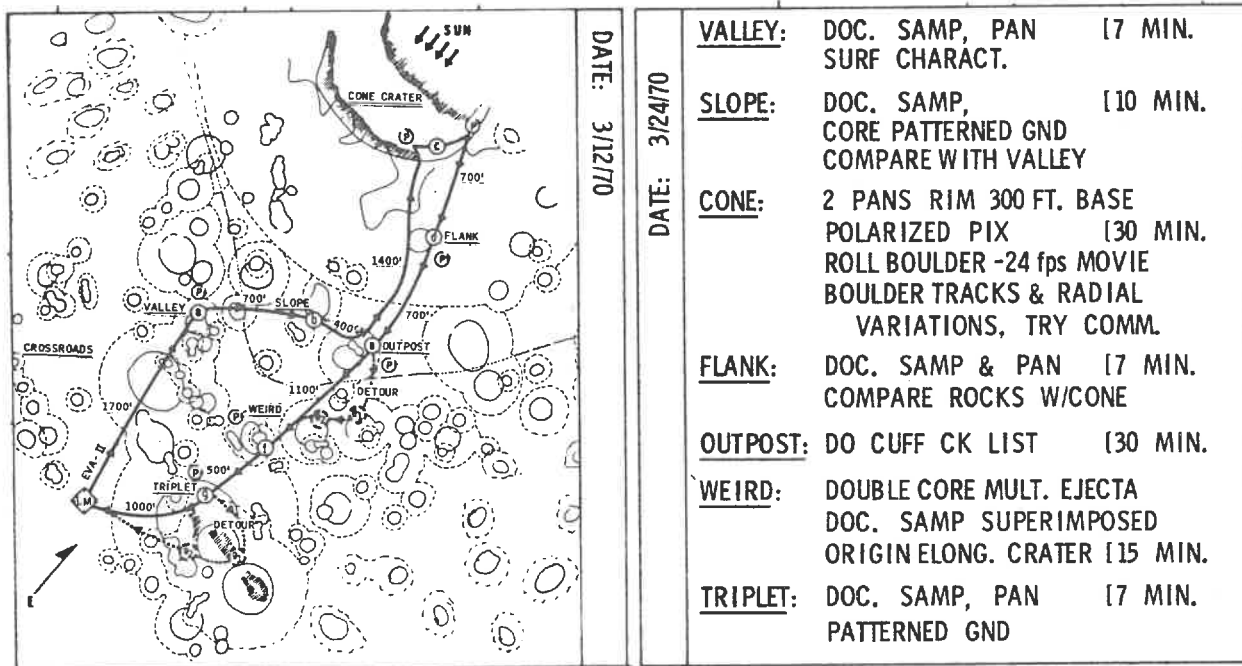
<u>TRAVERSE</u> <u>REPORT:</u> ALL MOVEMENTS INCLUDING DIRECTIONS BETWEEN SAMPLES LOCATION WRT LM PHOTOS OTHER THAN NOMINAL SAMPLE BAG NUMBERS OBSTRUCTIONS WRT LM- GET BEHIND & TRY COMM (CDR) LMP PHOTO OBSTRUCTION, CDR, LM <u>PHOTOS</u> PAN AT EACH LEG, SITE, DIRECTION CHANGE--MAP SHOWS NOMINAL CSC CAMERA PICTURES-PIGGY BACK ON DOC. SAMP. AS REQUIRED <u>REPORT ORIENT & FRAME NO.</u> <u>CRATER DOCUMENTATION</u> SERIES OF SAMPLES RADIALY FROM CENTER	DATE: 3/12/70
--	----------------------

DATE: 3/12/70	<u>CORE SAMPLES (SINGLE & DOUBLE)</u> TAKE ON SURFACE ONLY(SEE MAP) BEFORE: REPORT S/N TUBE(S) DRIVE TUBE(S) & PHOTO X SUN 15 FT AFTER: PHOTO X SUN STEREO 5 FT <u>TRENCHES</u> PHOTO SITE X SUN, DOWN SUN DIG DOWN SUN OFF 10° f:5.6,1/125 DOCUMENTED SAMPLES BOTTOM-TOP FILL, BOTTOM, TOP, SIDES,DISCONS CSC CAMERA PHOTOS BOTTOM RESET 70 MM CAMERA 1/250 <u>COLLECT 1 LARGE ROCK- FOOTBALL</u> <u>USE COLOR CHART</u> <u>COLLECT FILLET MAT'L & ROCK</u> <u>DOCUMENT & USE CSC CAMERA</u> <u>SAMPLE ROCK TRAIL</u> <u>FINES IN & BESIDE-CHIP FROM ROCK</u> <u>USE CSC CAMERA</u> <u>SAMPLE CRATER RIM CRESTS</u>
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SUR-72

Basic Date _____

Changed _____



SUR-73

Basic Date _____ 3/23/70
Changed _____ 3/31/70

Basic Date 3/23/70
 Changed 3/31/70

OUTPOST	<u>OUTPOST</u> 1. COLLECT GAS ANALYSIS SAMPLE (SMALL CAN): SEVERAL CLEAN ROCKS, GLASSES, SPLATTERED ROCKS, VARIETY DESIRED, DOCUMENT COLLECTION AS USUAL 2. COLLECT MAGNETIC SAMPLE (TEFLON CAN) 2-3 ROCKS MICROBRECCIA & CRYSTALLINE DOCUMENT COLLECT AS USUAL 3. DIG DEEP TRENCH (SOIL MECH.) DOCUMENT BEFORE. THEN f:5.6, 1/125 DIG 2 FT. DEEP 10° OFF DWN SUN DOCUMENT BEFORE SAMPLING: LMP STAND DWN SUN EDGE TRENCH CDR STEREO PR CROSS SUN BOTH SIDES, THEN 1 SHOT DWN SUN LMP 1 SHOT UP SUN	DATE: 3/25/70
DATE: 3/12/70	COMMENT ON SOIL PHYSICAL, MECH PROPS TAKE DOC. SAMPLES BOTTOM TO TOP, SIDES, DISCONS, FILL. CSC CAMERA IN BOTTOM, SIDES DOCUMENT BOOT PRINT IN FILL 4. COLLECT SPEC. ENVIRON SAMPLE (LARGE CAN): FINE-BOTTOM DEEP TRENCH - DOCUMENT COLLECTION 5. (LMP) a. USE CSC CAMERA b. RADIAL FINE SAMPLES SW OUTPOST CRATER - 3 @ 10 FT. INTERVALS - PHOTO AFTER 6. (CDR) GO BEHIND ROCK OR INTO CRATER TO TEST COMM. 7. USE COLOR CHART WITH LARGE ROCKS, COLOR CONTRASTS 8. COLLECT FOOTBALL	OUTPOST

SUR-74

Basic Date _____

<u>CSC CAMERA TIPS</u> REPORT SUN COMPASS/MAINTAIN STD ORIENTATION. REPORT FRAME NO. TAKE: UNEXPECTED FEATURES (TAKE SAMPLE) GLASSY FEATURES (TAKE SAMPLE) ROCK/SOIL JUNCTION-UP/DOWN HILL (TAKE SAMPLE ROCK & FILLETS) UNDISTURBED SURFACES (LEVEL & SLOPE) SURFACE PATTERNS -STRIATIONS ROCK SURFACES-FISSURES,CLEAVAGES, BEDDING, CRYSTALLIZATION, COLOR CONTRASTS SMALL CRATERS ON ROCKS & SURFACES "LEE" OF ROCKS (FROM ROCKET BLAST)	3/12/70
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DATE: 3/12/70	3+20 EVA CLOSEOUT ◦ REPOSITION TV OFF + Y ◦ [REMOVE TOTE BAG ◦ HOLD TOTE BAG FOR OFFLOAD ◦ TAKE OFF CDR TOTE BAG & HOLD FOR OFFLOAD ◦ HAND HTC SAMPLES TO CDR ◦ CSC CAM MAG IN ETB ◦ TAKE DOWN SOLAR WIND - BAG & PUT IN ETB ◦ CAMERA IN ETB ◦ SPARE MAG IN ETB ◦ SEQ CAMERA MAGS (3) IN ETB ◦ MAP IN ETB ◦ COLLECT EXTRA BAG 50 FT FROM LM ◦ SEAL & STOW IN ETB OR TOTE BAG
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SUR-75

Basic Date 3/23/70

Changed _____

Basic Date 3/23/70
Changed _____

CDR & LMP CUFF
CHECKLIST

TAKE OFF TONGS	
3+40 <u>EVA TERMINATION</u> CLEAN EMU INGRESS HOOKUP LEC	[TRANS PREP
3+47 <u>LEC TRANSFERS</u> STOW ETB STOW TOTE BAG STOW SRC 2	[ASCENT
3+52 HAND OUT LEC TO CDR	[INGRESS
4+00 REPRESS	

DATE: 3/12/70

PAGE 1 OF 5	EMU MALFUNCTIONS
	<u>EMU 1: VENT/TONE [P]</u> FAN OFF/ON IF FLAG STILL ON AFTER 10 SEC, ACTUATE OPS AND PURGE VLV
	<u>EMU 2: PRESS/TONE [O]</u> ACTUATE OPS IF FLAG ON, CK. CUFF GAGE, VER. TM; OPS OFF (SEN. FAIL)
	<u>EMU 3: O₂/TONE [O]</u> CK. LOW/HIGH PGA PRESS OR PLSS O ₂ DECREASING IF LEAK CONFIRMED, ACT. OPS IF HIGH PGA PRESS, ACT. OPS, PLSS O ₂ OFF (PLSS REG FAIL) IF NO APPARENT FAIL, VER. TM (SEN. FAIL)
	DATE: 3/12/70

SUR-76

Basic Date _____

EMU 4: H2O/TONE [A]

VERIFY FEEDWATER OPEN

IF CLOSED: DIV VLV MIN COOL,
FEEDWATER OPEN, WAIT 4 MIN.

IF ADDITIONAL COOL REQ'D, ACTUATE
OPS AND PURGE VLV

EMU 5: TONE/NO FLAGS

CUFF GAGE: IF < 3.4, ACTUATE OPS

IF CUFF GAGE > 3.4 & TONE OFF:
CYCLE MODE SW MOMENTARILY -
NO TONE, NO FAILURE.

IF TONE ON AGAIN:

FAN OFF 5 SEC, VENT FLAG [P]
SHOULD COME ON; FAN ON. IF
NO FLAG, ACTUATE OPS AND PURGE
VLV (FLAG FAIL)

DATE: 3/12/70

PAGE 2 OF 5

EMU 5: (CONT)

OBSERVE PLSS O₂: IF DECREASING,
ACTUATE OPS.

IS FEEDWATER CLOSED: IF CLSD,
DIV VLV MIN COOL, FEEDWATER
OPEN. AFTER 4 MIN, IF COOL
INADEQUATE, ACTUATE OPS AND
PURGE VLV.

EMU 6: CUFF GAGE < 3.7 - NO FLAG

ACTUATE OPS: CUFF GAGE SHOULD
RISE (EMU LEAK OR REG. SHIFT)
IF NO GAGE RESPONSE, VERIFY TM -
OPS OFF (GAGE FAIL)

PAGE 3 OF 5

DATE: 3/12/70

SUR-77

Basic Date _____
Changed _____

DATE: 3/12/70

PAGE 4 OF 5

EMU 7: PLSS O₂ QUANTITY ABNORMAL

- CK CUFF GAGE OR O₂ FLAG [O]
- IF LEAK CONFIRMED, ACT OPS
- IF HIGH PGA PRESS, ACT OPS,
PLSS O₂ OFF (PLSS REG FAIL)
- IF NO APPARENT FAIL, VERIFY TM
(GAGE FAIL)

EMU 8: CUFF GAGE > 4.0

- IF O₂ FLAG [O], OR PLSS O₂ DEC.,
ACT OPS AND PLSS O₂ OFF (PLSS
REG FAIL)
- IF NOT, VERIFY TM (POSS. GAGE FAIL)

EMU 9: LOSS OF PUMP NOISE

- IF NO SIDETONE, ACT OPS AND
PURGE VALVE (POWER FAIL)
- IF SIDETONE OK: CHECK PUMP ON,
VERIFY TM FOR PUMP CONDITION
- IF ADDITIONAL COOLING REQUIRED,
ACTUATE OPS AND PURGE VALVE

PAGE 5 OF 5

DATE: 3/12/70

EMU 10: COOLING INADEQUATE

VERIFY DIV VLV MAX
ACTUATE GAS TRAP FOR 5 SEC.,
IF FEASIBLE
AFTER 3 MIN., IF ADDITIONAL
COOLING REQUIRED, ACTUATE OPS
AND PURGE VLV

EMU 11: LOSS OF VOICE COMMUNICATION

CHECK VOL CONTROLS (WHEEL A,
BLADE B)
CYCLE PTT SW - MAIN & MOM
CDR MODE SW TO B, LMP TO A
(HAND SIGNALS)
IF NO COMM, CDR TO A, LMP TO B

Basic Date 3/23/70
Changed 3/31/70

POST EVA 2

PLSS FEEDWATER - CLOSE
Fwd Hatch - Close & Lock
Dump Valves (Both) - AUTO

Note: PLSS O2 & PRESS Flags May Come
On During Repress. If PLSS O2 <10%
Manually Control Cabin Repress To
Maintain Positive PGA Pressure.
(Leave Cabin Repress CB Open For
Manual Repress)

Lighting: ANUN/NUM - BRIGHT

CABIN REPRESS VLV - AUTO
CB(16)ECS: CABIN REPRESS - Close
(MASTER ALARM & Cabin Warning
Lt - On)
Verify Cabin Press Increasing
PRESS REG A & B - CABIN

PLSS O2 - OFF @ Cabin > 2.5 Psia
CABIN REPRESS Valve Closes At 4.4 Psia
Verify Cabin Press Stable At 4.6
to 5.0 Psia (Cabin Warning Lt - Off)

Operate OPS Purge Valve To Depress
Suit As Required
DET - Set/Up :00

POST EVA SYSTEMS CONFIGURATION :00

Verify EVA CB Configuration
CB(11) ECS: SUIT FAN 1 - Close
CB(16) ECS: SUIT FAN ΔP - Close
(ECS Caution & H2O SEP Comp Lts - Out)

Doff Gloves

DES H2O VLV - OPEN
Remove Purge Valve & OPS O2 Hose
Stow Purge Valves in Purse

Connect LM O2 Hoses (R/R & B/B)
SUIT ISOL (Both) - SUIT FLOW

PLSS PUMP - OFF
PLSS FAN - OFF

SUR-79

EVA 2 POST

Disconnect PLSS H2O From PGA
Connect LM H2O To PGA
CB(16) ECS: LCG PUMP - Close

PLSS Mode (Both) - 0
Connect To LM Comm (Audio, Biomed)

AUDIO (CDR & LMP)

VHF A - OFF
VHF B - OFF
MODE - ICS/PTT
RELAY - OFF

COMM:

VHF - OFF,OFF,OFF,OFF,LEFT,HI
RECORDER - OFF

PLSS/OPS DOFFING :10

Disconnect OPS Actuator From RCU
Disconnect RCU From PGA
Verify All RCU Controls - OFF
Disconnect RCU From PLSS And Stow
.On Mid-Step

Disconnect PLSS 02 Hoses
Doff PLSS/OPS
Stow LMP PLSS On Floor
Stow CDR PLSS On Mid-Step

Install Gas Connector Plugs (Purse)
In PGA

Stow OPS 02 Hoses, Actuator, & Antenna
Disconnect OPS Antenna Connector
Doff Yo-Yos, Stow In LHSSC
Stow PLSS Hose Nozzles

Remove OPS & Stow Antenna Connector
Perform OPS Checkout (Both)
Stow OPS On Engine Cover, End Up
Remove Lower PLSS Straps, Clip
Straps Together (Name-To-Name)
& Stow In RHSSC(FECAL EMESIS)
Stow PLSS (Both) On Floor

Verify Powerdown CB Configuration
CB(11) HEATERS: RR OPR - Close

SUR-80

Basic Date 3/23/70
Changed _____

Basic Date 3/23/70
Changed 4/3/70

PREP FOR EQUIPMENT JETTISON :37

Verify 02 QTY > 25%

Fwd Hatch Handle - UNLOCK

Unstow Jett Bag, Aft LHSSC,

Upper RH Corner

Doff Lunar Boots, Stow In Jett Bag

Empty ETB On Eng Cover

Stow ETB In Jett Bag

Stow Unused 16mm Mag (RHSSC) In Purse

Stow 70mm Mags(2) & 16mm Mag(3) In RHSSC

Stow 70mm Mag (1) In Bag (Top ISA) &

Stow Bag In Lower Hasselblad Compt

Stow Return Items In ISA

Stow 70mm Camr(1), Triggers(2), & Handles
(2) In Jett Bag

Reconfigure 70mm Camr(1) & Stow In RHSSC

Verify 70mm(5) & 16mm (8) Mags

Stow Butterflies & Solar Wind In ISA

Stow Sieve & Extra Sample In Tote Bag #2

Stow CSC Cassette In Upper Boot Compt

Remove PLSS Condensate Container, + **CONTAM**
Stow In Jett Bag **SAMPLE**

Stow In LHSSC:

RCU's (2)

Unused Def Bags As Desired

Yo-Yos (4)

Food Waste

Urine Bags

Report SRC & Tote Bag #2 Wt
To Hou (Scale Bottom ISA)

Stow Scale In Jett Bag

Lower CDR RH Attitude Controller

Remove Armrest, Stow In Jett Bag

Position PLSS (2) For Jettison, Eng
Cover & Mid-Step

Unsnap LHSSC For Jettison - Leave 1 Snap

Tie Jett Bag & Transfer To Eng Cover

SUR-81

Basic Date

Basic Date _____
Changed _____

Don EV Gloves

PGA Diverter Vlv - Horizontal
Check PGA Connectors

PRESS INTEGRITY CHECK :57

Note: ARS/PGA Circuit Shall Not Be
Maintained At Elevated Press >5 Min

SUIT GAS DIVERTER - PULL-EGRESS (Verify)
CABIN GAS RETURN - EGRESS (Verify)
SUIT CIRCUIT RELIEF - CLOSE

PRESS REG A - CLOSE
PRESS REG B - DIRECT 02
Monitor Suit Press To 8.85 Psia
Then PRESS REG B - CLOSE (Cuff
Gage Decay <.3 Psig In 1 Min)

SUIT CIRCUIT RELIEF - AUTO (Suit
Press Decays To 4.8 Psia)
PRESS REG A & B - EGRESS

CABIN DEPRESS FOR JETTISON 1:00

CB(16)ECS: CABIN REPRESS -Open
Fwd Dump Valve - OPEN Then AUTO At
3.5 Psia
(Verify Cabin Press 3.5 Psia
& LM Suit Circuit 3.6 To 4.3
Psia & Decaying)

Fwd Dump Valve - OPEN (Verify LM
Suit Circuit 3.6 To 4.3 Psia)

HATCH OPENING 1:03

Partially Open Fwd Hatch
Fwd Dump Valve - AUTO

Fwd Hatch - Full Open

Jettison The Following:
PLSS On Mid-Step
Jett Bag
PLSS On Eng Cover
LHSSC

SUR-82

Basic Date 3/23/70
Changed _____

Basic Date 3/23/70
Changed _____

Verify Items Clear Of Ascent Stage

Fwd Hatch - Close & Lock

CABIN REPRESS 1:07

Fwd Dump Valve - AUTO (Verify)

CABIN REPRESS - AUTO (Verify)
CB(16)ECS: CABIN REPRESS - Close
(MASTER ALARM & Cabin Warning
Lt - On)

Verify Press Increasing
PRESS REG A & B - CABIN

CABIN REPRESS Valve Closes At 4.4 Psia
Verify Cabin Press Stable At 4.6 To 5.0
Psia (Cabin Warning Lt - Off)

CABIN GAS RETURN - AUTO
SUIT GAS DIVERter - PUSH-CABIN

Doff Gloves, Helmets With Visors
VHF ANT SEL - AFT

Unstow Lunar Surface Checklist (SUR-83)
Stow EVA 2 Prep & Post Card

POST EVA CABIN CLEANUP 1:11

Secure OPS (2) On Floor

Stow Cuff Checklist In ISA
Stow EV Gloves & Visors In Helmet Bags

Stow Helmet Bags On Floor

Stow Purge Valves (Purse) in
RHSSC (FECAL EMESIS)

Stow SRC

Stow LM EVA Antenna

Stow Return Items In ISA
Snap Tote Bags #1 & 2
Remove ISA
Stow Tote Bag #2 (#1 If Possible)
Inside ISA
Stow Tote Bag #1 In Cover Bag
(Back ISA) If Required
Secure Cover Bag Or Tote Bag To
Back of ISA

SUR-83

Basic Date _____
Changed _____

Changed _____

Install ISA In Rest Station
Report Tote Bag #2 Wt _____ To Houston
(Scale ISA)

Copy P22 ACQ Time
_____:____:____

Secure Helmet Bags On Engine Cover
Secure Utility Lights On AOT

Stow All EVA Onboard Data
In Flt Data File

DET-Stop 1:30

EVA 2 DEBRIEFING WITH HOU (5 MIN)

EAT PERIOD <u>133:20 TO 134:20</u>

Copy Liftoff Time In Data Book
For T - n

CREW STATUS REPORT

CDR

LMP

MED _____

PRD _____

LM CONSUMABLES UPDATE

GET (133:30) _____:

RCS A % (77) _____ B (75) _____

O2 DES % (38.2) _____ ASC (100) _____

H2O DES % (33.5) _____ ACS (100) _____

A-H DES (582) _____ ASC (566) _____

SUR-84

Basic Date 3/23/70

Changed _____

LM-7

Basic Date 3/23/70
Changed _____LAUNCH PREP***** LO-2:50 (134:19) *****

Verify:

MASTER ARM - OFF

GUID CONT - PGNS

ENG ARM - OFF

ATTITUDE CONT (3) - MODE CONT

MODE CONT (Both) - ATT HOLD

S-BAND-PM, PRIM, PRIM, VOICE, PCM,
OFF/RESET

VHF - OFF, OFF, OFF, OFF, LEFT, HI

POWER/TEMP MON - Check BAT, BUS Volts

PRO

(RESTART Lt-On, STBY Lt-Off)

RSET

V96E

CB(11) IMU OPR - Close

(NO ATT Lt-Off In 90 Sec)

CB(11) AGS - Close

AGS STATUS - OPERATE

(AGS Warn Lt-On)

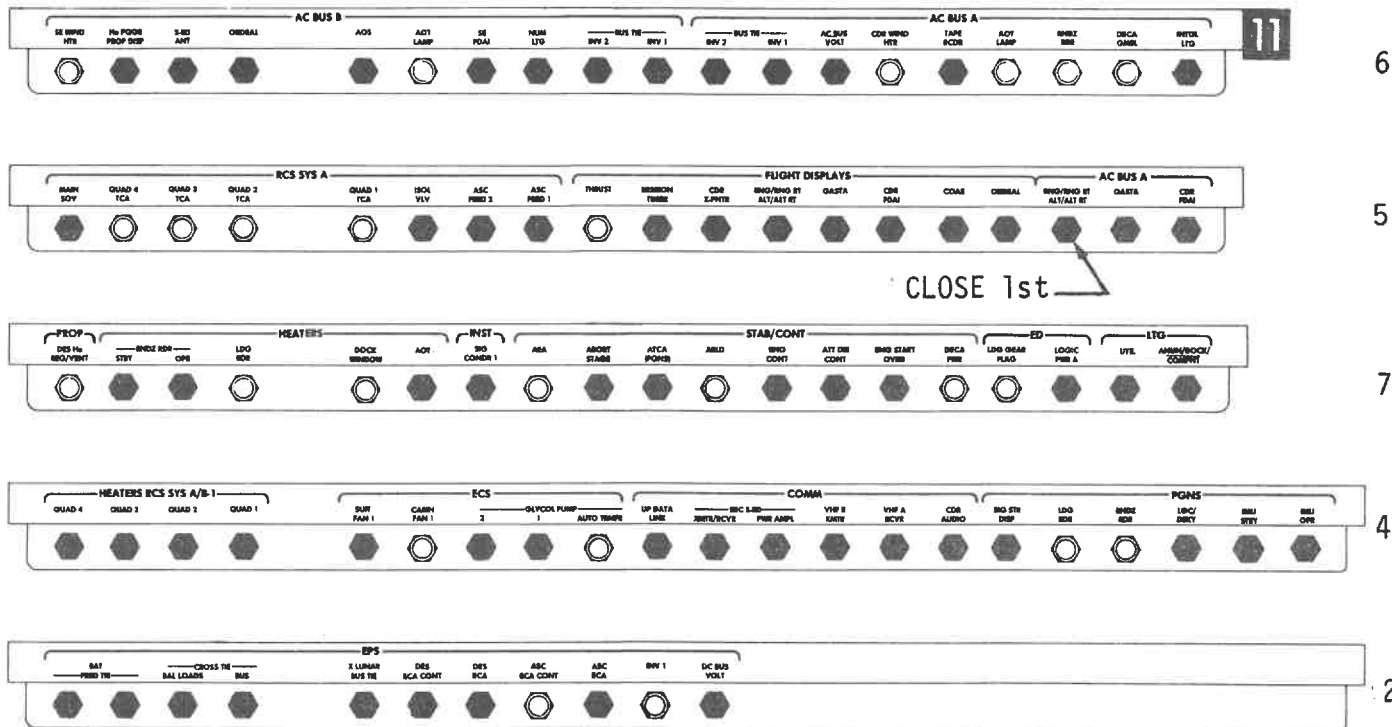
Configure CB's Per PWR UP Chart

SUR-85

LAUNCH PREP

LAUNCH PREP

PWR UP



SUR-86

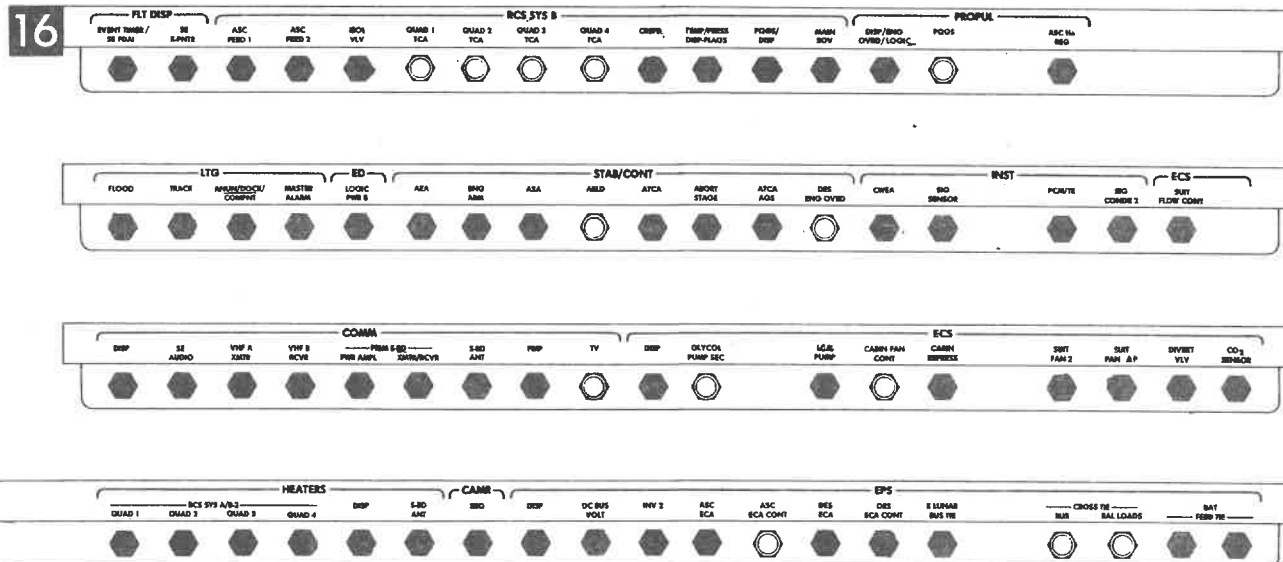
LM-7

Basic Date 3/23/70
 Changed _____

LM-7

Basic Date 3/23/70
 Changed _____

PWR UP



SUR-87

Basic Date _____
 Changed _____

CB(16) Cycle CWEA

000 + 888888 (OPR ERR Lt-On)

V25 N01E, 1365E

123 - 45679

E,E,E

412R + 1 Satisfactory

V15 N01E, 1365E

574R + Not Staged

V21 N27E, 10E

604R - On Surface

15 01 Test Successful When
R2 >3 (78 sec)

612R + 0 ATT HOLD At
ABORT STAGE

V21 N27E, 0E

Notify MSFN of E-Dump

TLM-HI

V74E (42 sec)

***** LO-2:40 (134:29) *****

DET - Set Counting Down To Acq Time
UP DATA LINK-DATA (MSFN Uplinks
CSM State Vector & Zeroes POS/NEG
Cells), OFF

SUR-88

LM-7

Basic Date 3/23/70
Changed

LM-7

Basic Date 3/23/70
Changed

***** L0-2:35 (134:34) *****

CB(11) RR (2) - Close
V41N72E (00000,28300)
V16N72E
CB(11) RR (2) - Open
V44E

CB(11) AOT-Close
P57, SET R2 00004
N34 (Load L0 Time), PRO
N06 00010
 00003
 00010
PRO
(NO ATT Lt - On/Off, Twice)
N04 + ΔTilt (.01°)
 V32E(Recycle)
N04 ΔTilt (.01°)
PRO
N22 ICDU Angles
PRO (NO ATT Lt - On/Off)
STAR ID (SHAULA 200)
PRO ,PRO
 Cursor
 Spiral
N79 Load Then V32E
 Cursor
 Spiral
N79 Load Then V32E

SUR-89

Basic Date
Changed

Cursor _____
 Spiral _____
 N79 Load Then PRO
 N88 Load Vectors For SHAULA Then PRO
 X _____ (-.04936)
 Y _____ (-.39581)
 Z _____ (-.30149)
 N05 _____ Star Angle Diff (.01°)
 PRO _____
 N93 _____ X Torquing Angle (.001°)
 _____ Y
 _____ Z
 PRO (Gyro Torquing)
 N25 00014, ENTER (TERM)
 POOE
 CB(11) AOT - Open

CB(11) RR (2) - Close
 X-POINTER SCALE (Both) - HI MULT
 RATE/ERR MON (Both) - RNDZ RDR
 ATTITUDE MON (Both) - PGNS
 MODE SEL - AGS
 RNG/ALT MON - RNG/RNG RT
 SHFT/TRUN - +50°
 TEMP MONITOR - RNDZ (+10° To +75°)
 RR MODE - AUTO TRACK
 RADAR TEST - RNDZ (Rng Rt Tape
 Drives, X-POINTERS &
 FDAI Needles Vary +5°, After 12
 sec Rng Tape Drives NO TRACK Lt - Off)

V40N20E
 400 + 3 AGS/PGNS Align
 544R _____ X Gyro Coeff
 545R _____ Y Gyro Coeff
 546R _____ Z Gyro Coeff

400 + 6E Calibrate Gyros
 Wait 5 min 2 sec
 400R (+0 Calib Complete)

SUR-90

LM-7

Basic Date 3/23/70
 Changed _____

LM-7

Basic Date 3/23/70
Changed _____

TEST/MONITOR - AGC 1.0 To 1.8 (1.6)
- XMTR PWR 2.1 To 4.1 (2.4)
- SHAFT ERR 2.0 To 2.8
- TRUN ERR 2.1 To 2.7
- AGC

SET NORRMON Flag

V25 N07E

101E, 10E, 1E

RR MODE - LGC (NO TRACK Lt - On)

V63E, R2 00001, PRO

(NO TRACK Lt - Out After 12 sec)

N72 Varying @ 1/2 cps,

PRO

N78 +195.39 To +195.79 Rng (TM Within +1.2 of R1)
-00475 To -00517 Rng Rt (TM = R2 < 2)

V34E ,POOE

RADAR TEST - OFF

(NO TRACK Lt - On, X-POINTERS Center)

V40 N72E

V41N72E (18000, 27000)

V16 N72E

V44E

544R _____ X
545R _____ Y
546R _____ Z

If Gyro Coeff Changes More
Than 2.0°/hr, AGS Failed

V40N20E

400 + 3 AGS/PGNS Align

SUR-91

Basic Date _____
Changed _____

***** LO-2:10 (134:59) *****

V95E
P22E
N06 R2 00001
V83E, Rng <400, PRO, PRO
V16N72E (18000,33500)
(N78E Rng, Rng Rt)
At End of CSM Track:
N72 Goes To 18000, 27000
P00E

V41 N72E (00000, 28300)
V16 N72E
CB(11) RR(2) - Open
V44E

RATE GYRO CHECK
GYRO TEST - POS RT (RPY RATE +5°/SEC)
GYRO TEST - NEG RT (RPY RATE -5°/SEC)

RATE SCALE -5°/SEC
Repeat Tests

V16 N65E LGC TIME
377 (Bias 130)
V47 Set Bias, 414+1

P ., Y (+110/-41)
S-BAND ANT-SLEW (>3.0)
TRACK MODE-AUTO (>4.0)

***** LO-1:30 (135:39) *****

V48, 11102, PRO, V34E
V77E

Copy Ascent & CSI Pad

V15N01E,42E(Rate Cmd Hot Fire Check ACA to Copy LO DAP Pad
SUR-92 Jets)

LM-7

Basic Date 3/23/70
Changed

LM-7

Basic Date 3/23/70
 Changed _____

CB(11 & 16) QUAD TCA (8)-Close	224 _____	Low Lim (#58572)
CDR ACA (Out-Of-Detent, Pause 2 sec At	226 _____	Retarget Value For α L (#58572)
Roll Rt 000XX Null)	231 _____	RLS (#56978)
Lt 777XX	232 <u>+00600</u>	INS ALT
Pitch Up 000XX	410 <u>+0</u>	ORB INS
Dn 777XX	465 <u>+</u>	INS HDOT (+00320)
Yaw Rt 777XX	547 <u>+0</u>	LUNAR ALIGN CORRECTION
Lt 000XX	662R <u>+ 0</u>	(4K10)
CB(11&16) QUAD TCA (8)-Open	673R <u>+ 0</u>	
V76E(Min Imp Check of CDR ACA To LGC, ACA	514 R _____	(-60000)
Cold Fire CES Voltage, SEC RCS Coil Hot	515 R _____	(-44223)
Fire 4-JET In AGS)	516 R _____	(+00000)
V11 N10E, 31E, R1 67777		
GUID CONT-AGS	275 _____	TIG TPI (+0585.7)
ATTITUDE CONT (3)-MODE CONTROL	373 _____	TIG CSI (+0486.0)
CDR ACA (Deflect Slowly To Hardover, Pause	411+1	ASC ENG
2 sec At Null)	416+1	1/2 PERIOD
Roll Rt R1 27757	451+0	Δ VY
Lt 27737	605 <u>+00777</u>	COT LOS TPI
Pitch Up 27776		
Dn 27775		
Yaw Rt 27767 (QUAD Flag & RCS		
Lt 27773 TCA Warn Lt-On)		
GUID CONT - PGNS		
MODE CONT (AGS) - AUTO		

Cycle CWEA

SUR-93

Basic Date _____
 Changed _____

Changed _____

***** L0-:45 (136:24) *****

CB AOT LAMP - Close

Window Shades - Up

P57E, R2 00004

N34 Load TIG, PRO

N06 00010

00003

00110

PRO

(NO ATT Lt - On/Off, Twice)

N04 + _____ Δ Tilt (.01°)

V32E

N04 _____ Δ Tilt

PRO

N22 ICDU Angles

PRO (NO ATT Lt - On/Off)

STAR ID _____ (SHAULA 200)

PRO ,PRO

Cursor _____

Spiral _____

N79 Load Then V32E

Cursor _____

Spiral _____

N79 Load Then V32E

Cursor _____

Spiral _____

N79 Load Then PRO

SUR-94

LM-7

Basic Date _____ 3/23/70

Changed _____

LM-7

Basic Date 3/23/70
Changed

N88 Load Vectors For SHAULA Then PRO

X (-.04936)

Y (-.39581)

Z (-.30149)

N05 Star Angle Diff (.01°)

PRO

N93 Torquing Angles (.001°)

PRO (GYRO TORQUING)

N25 00014, ENTR (TERM)

POOE

Window Shades - Down

CB(11) RR(2) - Close

V41N72E (00000,00000)

V16N72E

CB(11) RR (2) - Open

V44E

V40N20E

400+3 AGS/PGNS Align

413+1

047R Transmit To MSFN

053R Transmit To MSFN

AOT - CL/0.0°

CB(11) AOT LAMP - Open

BAT 5,6 - ON

BAT 1,3 - OFF/RESET, tb-bp

CB(11&16) ASC ECA CONT(2)-Close

Don Helmets, Gloves, & Restraints

Set Camr For Ascent

(LM3/DAC/10/CEX ULC (f2.8,500,∞) 1 mag, 12fps,8min)

SUR-95

Basic Date
Changed

***** LO-:35 (136:34) *****

V48E

N46 12002 PRO

N47 _____ (10775) LM Wt
 _____ (36352) CSM Wt

PRO

GUID CONT - PGNS

UPDATA LINK - DATA
 (Possible MSFN Update Of
 CSM S.V., RLS and LGC Gyro
 Compensation)
 UPDATA LINK - OFF

P12E

N33 ____:____:____ TIG (137:09:16.00)

PRO

N76 _____ VH Final (+55349)

_____ HDot Final (+00322)

_____ Xrng (00000)

PRO

N25 R1 00203 (MODE CONT:PGNS - ATT HOLD)

ENTR

DET - Set/UP

SUR-96

LM-7

Basic Date 3/23/70

Changed _____

Basic Date 3/23/70

Changed _____

***** LO-:30 (136:39) *****

400 + 4E Lunar Align

CB(11) QUAD 4,3,2,1 TCA (4) - Close

DES He REG/VENT - Close

AELD - Close

INV 1 - Close

CB(16) QUAD 1,2,3,4 TCA (4) - Close

AELD - Close

MASTER ARM - OFF

STAGE - SAFE/Guarded

X POINTER SCALE (2) - HI MULT

RATE/ERR MON (2) - LDG RDR/CMPTR

ATT MON - PGNS

GUID CONT - PGNS

MODE SEL - AGS

RNG/ALT MON - ALT/ALT RT

RATE SCALE - 25°/SEC

ACA PROP (2) - ENABLE

ENG ARM - OFF

ATT/TRANSL - 4 JETS

BAL CPL - ON

ASC He REG 1&2 tb(2) -gray

ABORT - Reset

ABORT STAGE - Reset

ENGINE STOP (2) - Reset

PRPLNT TEMP/PRESS - ASC

SUR-97

Basic Date _____

Changed _____

Changed

HELIUM MON - ASC PRESS 2
SYS A&B QUAD 1,2,3,4, (8) tb-gray
SYS A&B ASC FUEL & OXID tb(4)-bp
SYS A&B MAIN SOV tb(2)-gray
CRSFD tb-bp
TEMP/PRESS MON - OXID MANF
GLYCOL - PUMP 1
SUIT FAN - 1
O2/H2O QTY MON - ASC 1
ATTITUDE MON - AGS
RADAR TEST - OFF
RR MODE - LGC
DEAD BAND - MIN
ATTITUDE CONTROL (3) - MODE CONT
MODE CONTROL (Both) - AUTO
TEMP MONITOR - RNDZ RDR
RCS SYS A/B-2 QUAD 1,2,3,4 - AUTO
ACA/4 JET (Both) - ENABLE
TTCA/TRANSL (Both) - ENABLE
TTCA (Both) - JETS (Dn)

DES H2O - CLOSE
ASC H2O - OPEN
WATER TANK SEL - ASC
CABIN REPRESS - CLOSE
DES O2 - CLOSE
ASC O2 No. 1 - OPEN

SUR-98

Basic Date 3/23/70

Changed

LM-7

Basic Date 3/23/70
Changed _____

PRESS REG A&B - EGRESS
SUIT GAS DIVERTER - PULL/EGRESS
CABIN GAS RETURN - EGRESS
SUIT CIRCUIT RELIEF - AUTO

DES FUEL & OXID VENTS (2) - OPEN
(tb(2)-gray)
DES He REG 1&2 - OPEN
(tb-gray)
CB(11) DES HE REG/VENT - OPEN
ASC He REG 1&2 - tb(2) -gray
MASTER ARM - ON
ASC He SEL - TANK 1
ASC He PRESS - FIRE
ASC He SEL - TANK 2
ASC He PRESS - FIRE
MASTER ARM - OFF

SYS A ASC FEED 2 - OPEN tb(2)-gray
Monitor SYS A Manf Press
SYS A MAIN SOV - CLOSE tb-bp

SYS B ASC FEED 2 - OPEN tb(2)-gray
Monitor SYS B Manf Press
SYS B MAIN SOV - CLOSE tb-bp
CRSFD - OPEN, tb-gray

SUR-99

Basic Date _____
Changed _____

***** LO-:17 (136:52) *****

V47E, 414 + 1
UPDATA LINK - VOICE/BU
VHF A: XMTR - VOICE/RNG (HOT MIKE TO CSM)
RCVR - OFF
VHF B: XMTR - OFF
RCVR - ON.
AUDIO (Both) VHF A - T/R
VHF B - RCV
RECORDER - ON
MODE - ICS/PTT
VHF ANT - AFT
GO For LO & Guidance
Recommendation From MSFN

***** LO-:15 (136:54) *****

BAT 2,4 - OFF RESET, tb-bp
DES BATS - DEADFACE, tb-bp
If tb-bp,
CB(11 & 16) DES ECA - Open
CB(11 & 16) DES ECA CONT - Open

Verify CB's Per LAUNCH Configuration Chart
(Next 2 pages)

SUR-100

LM-7

Basic Date 3/23/70
Changed

Basic Date 3/23/70
Changed _____

[illegible][illegible][illegible][illegible]

BAT
 FUSE TIE
 CHARGE TIE
 BAL. LOADS
 BUS
 EPS
 E LAMP
 BUS TIE
 RES
 BGA CONV
 RES
 BGA
 ARC
 BGA CONV
 RES
 BGA
 INV 1
 RES
 BGA
 VOLT

SUR-101

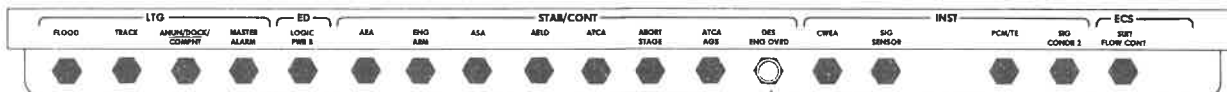
Basic Date _____
Changed _____

LAUNCH

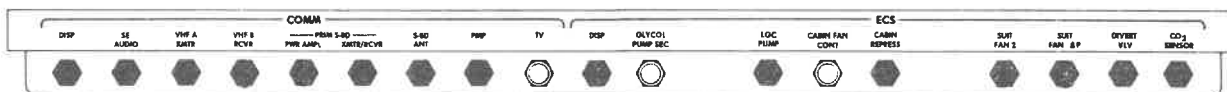
16



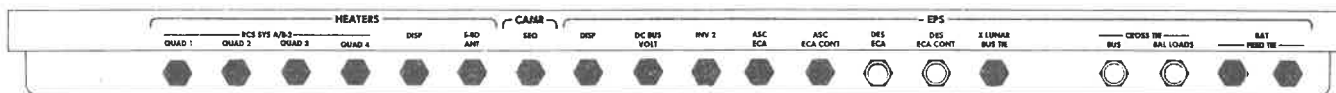
1



1



3



4

SUR-102

LM-7

Basic Date 3/23/70
 Changed _____

LM-7

Basic Date 3/23/70
Changed _____

***** L0-:10 (136:59) *****

CHECK APS BURN CARD

***** L0-:5 (137:04) *****

CHECK APS, RCS, EPS, ECS

GO TO LM TIMELINE BOOK

SUR-103

Basic Date _____
Changed _____

Basic Date 3/23/70
Changed _____

ONE MAN EVA PREP

CABIN PREP-Perform EVA 1 Or 2 As Req'd

EQUIPMENT PREP-Perform EVA 1 Or 2 As
Required

PLSS DONNING-Perform EVA 1 or 2 As Req'd

Position Post EVA 1 or 2 Cue Cards
For Post EVA

Fwd Hatch Handle - UNLOCK

NON EVA CREWMAN-Connected To LM 02,
COMM, & H2O
Gas Connector Plugs In PGA

EVA CREWMAN: For EVA 1-
CSRC In PGA Pocket

PLSS COMM CHECK

Verify PWRDN CB Configuration
Verify LM EVA Antenna Deployed
COMM: MODULATE-FM
PWR AMPL - PRIM

CB(16) COMM: TV-Close
Verify Voice Comm With Hou

Audio (Non EVA Crewman)

S-BAND - T/R

ICS - T/R

RELAY - OFF

MODE - VOX (VOX SENS MAX)

VHF A - RCV

VHF B - T/R

Audio (EVA Crewman)

S-BAND - T/R

ICS - T/R

RELAY - ON

MODE - VOX (VOX SENS MAX)

VHF A - RCV

VHF B - T/R

COMM:

VHF-OFF,ON,VOICE,ON, NON EVA
CREWMAN POSITION, HI

SQUELCH A&B - Noise Thres + 1-1/2

RECORDER - ON

VHF Antenna - EVA

SUR-104

ONE MAN EVA PREP

EVA Crewman Connect to PLSS Comm
(Audio CB)

RCU PTT - MAIN

PLSS Mode-B (Tone-On, Vent Flag-P, Press
Flag-0)

PLSS 02 Press Gage >85% (75% for EVA 2)
Perform Comm Check With CDR

Note: Unstow PLSS Antenna If It Transmits
Garbled And/Or Loses TM.

Audio (Non EVA Crewman)
VHF A - T/R
VHF B - RCV

Audio (EVA Crewman)
VHF A - T/R
VHF B - RCV

COMM:
VHF A XMTR - VOICE
VHF B XMTR - OFF

PLSS Mode - A (Tone-On)
Perform Comm Check With Each Other &
Comm & TM Check With Hou

Read PLSS 02 Qty To Hou

Note: IF Comm Is NO GO With Hou
S-BD MOD - PM
Verify Comm & Tm

CB(16) COMM: TV - Open

FINAL SYSTEMS PREP

CB(16) ECS: CABIN REPRESS - Close (Ver)
SUIT FLOW CONT- Open

SUIT GAS DIVERTER - PULL-EGRESS
CABIN GAS RETURN-EGRESS
SUIT CIRCUIT RELIEF-AUTO (Verify)

OPS CONNECT

Unstow OPS 02 Hose & Actuator
Connect Actuator To RCU

SUR-105

Basic Date 3/23/70
Changed

Basic Date 3/23/70
Changed _____

Snap OPS 02 Hose To Side Of PLSS
SUIT ISOL - SUIT DISC
Discon LM 02 Hoses, Secure About PGA

Connect OPS 02 Hose To PGA B/B
Retrieve Purge Valve (Purse) -
 Verify Closed & Locked
Install Purge Valve In PGA R/R
Verify PLSS Centered & At Proper
 Height

FOR EVA 2:
 Verify Items Prepared For Jettison -
 ECS LiOH Cartridge - Jett Bag
 Hammocks - Jett Bag
 LHSSC (Fwd Section) - PLSS Batteries,
 LiOH Carts, Food Waste, Urine Bags
 Position ETB On Mid-Section Step

Drink
DES H2O VLV - CLOSE

HELMET/GLOVE DONNING

Position Mikes (Both)
PLSS FAN - ON (Vent Flag - Clear)
Don Helmets, Then Visors
Check Drink Bag Position
Unstow EV Gloves
Position Helmet Bags In SRC Area

EVA Crewman:
 LCG - COLD, As Req'd
 Disconnect LM H2O Hose
 Connect PLSS H2O Hose
 Stow LM Hoses

Verify EVA Crewman in CDR's Station

Verify The Following:
 Helmet & Visor (2) - Locked &
 Adjusted
 Torso Tiedown (2) - Adjusted
 02 Connectors (7) - Locked
 Purge Valves (1) - Locked
 H2O Connectors (2) - Locked
 Comm Connectors (2) - Locked

SUR-106

Basic Date _____

Basic Date _____
Changed _____

Verify No Fog RH Window
Tie Jett Bag & Transfer To Eng Cover

Don EV Gloves & Verify:
Wrist Locks (4) - Locked
Glove Straps (4) - Adjusted
If PGA Biting, PLSS O2 - ON/OFF

PLSS DIVERter - MIN (Verify)
PLSS PUMP - ON

PRESSURE INTEGRITY CHECK

ARS/PGA (Non EVA Crewman)

Note: ARS/PGA Circuit Shall Not Be
Maintained At Elevated Press >5 min

SUIT GAS DIVERter - PULL-EGRESS (Verify)
CABIN GAS RETURN-EGRESS (Verify)
SUIT CIRCUIT RELIEF - CLOSE

PRESS REG A - CLOSE
PRESS REG B - DIRECT O2
Monitor Suit Press To 8.85 Psia
Then PRESS REG B - CLOSE (Cuff Gage
Decay <.3 Psig In 1 min)

SUIT CIRCUIT RELIEF - AUTO (Suit Press
Decays To 4.8 Psia)
PRESS REG A & B - EGRESS

PLSS/OPS/PGA (EVA Crewman)

PLSS O2 - ON (Tone-On, O2 Flag-0)
Press Flag Clear (3.1-3.4 Psid)
Cuff Gage 3.7-4.0 Psig
O2 Flag Clear

PLSS O2 - OFF (Cuff Gage Decay <.3
Psig In 1 Min)

PLSS O2 - ON (Cuff Gage 3.7-4.0
Psig, Tone & O2 Flag May Come On)

SUR-107

Basic Date 3/23/70
Changed _____

Basic Date 3/23/70
Changed _____

CABIN DEPRESS

Confirm "Go" For Depress From Hou
CB(16)ECS: CABIN REPRESS - Open
CABIN REPRESS VLV - Close

Fwd Dump Valve - OPEN Then AUTO At
3.5 Psia (Verify EVA Crewman Cuff
Gage Does Not Drop Below 4.8 Psig)

Verify:

Cabin At 3.5 Psia
LM Suit Circuit 3.6 to 4.3 Psia &
Decaying
PLSS/OPS/PGA >4.8 Psig & Decaying

Start Wrist Watch :00

Fwd Dump Valve - OPEN

Verify:

Tone-On & H2O Flag-A(1.3-1.6 Psia)
LM Suit Circuit 3.6 To 4.3 Psia &
Decaying
PLSS/OPS/PGA >4.8 Psig & Decaying

Partially Open Fwd hatch

FINAL PREP FOR EGRESS :03

PLSS FEEDWATER - OPEN (H2O Flag -
Clear In About 4 Min)
Fwd Hatch - Open

Rest Until Cooling Sufficient
Verify:

PLSS/OPS/PGA 3.7 To 4.8 Psig
LM Suit Circuit 3.6 To 4.3 Psia
CWEA Status:

<u>Warning</u>	<u>Caution</u>
ASC PRESS	PREAMPS

CB(16) COMM: TV - Close

For EVA 1: Jettison At End of EVA 1
- Malfunctioned Equipment Which Is
NO-GO FOR EVT

For EVA 2: Jettison At Start of EVA 2
- Malfunctioned Equipment Which Is
NO-GO For EVT
- Jettison Bag & LHSSC

Release PLSS Antenna

SUR-108 Lower EV Visor :10

Basic Date _____
Changed _____

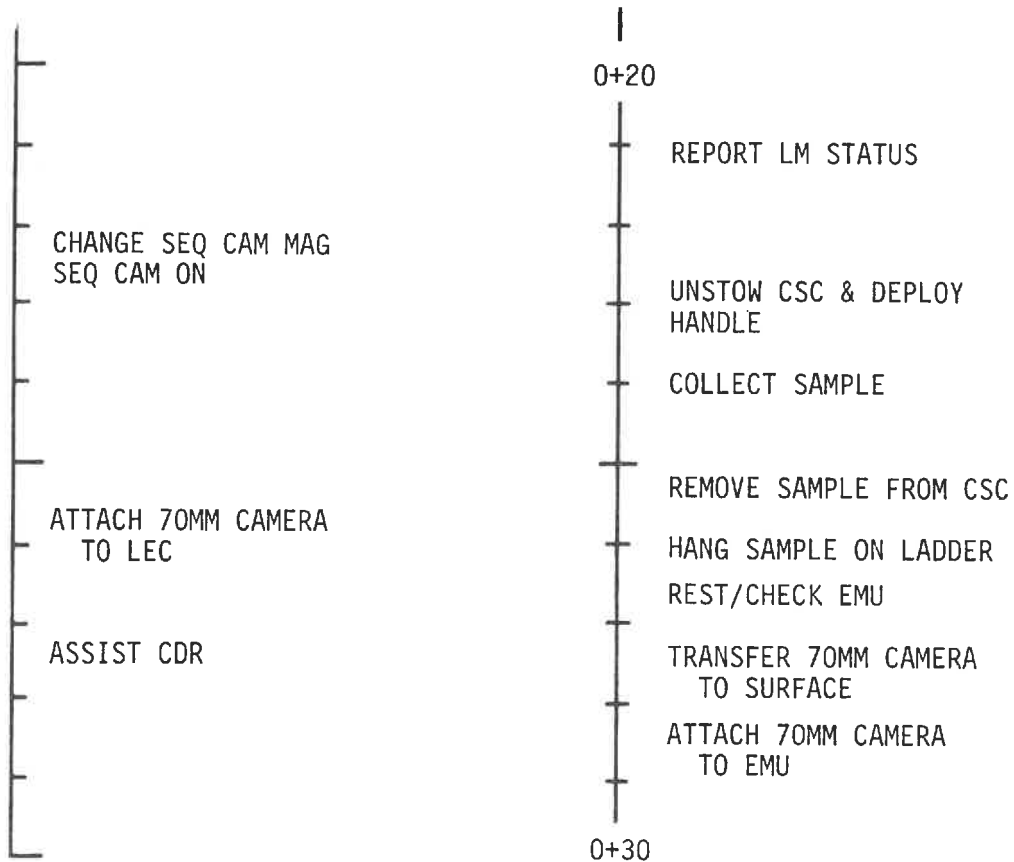
Basic Date 3/23/70
Changed _____

LMP ACTIVITIES	EVA TIME	CDR ACTIVITIES
	0+10	MOVE THROUGH HATCH
PREPARE LEC		CHECK INGRESS PROCEDURES
PASS LEC TO CDR		DEPLOY LEC
PHOTOGRAPH CDR		DESCEND TO LADDER DEPLOY MESA
SEQ CAM ON		DESCEND TO FOOTPAD
NOTE: • MONITOR & PHOTOGRAPHY CDR USING 70MM • READ PROCEDURES TO CDR		CHECK ASCENT PROCEDURES
		STEP TO SURFACE
		CHECK & DISCUSS MOBILITY & STABILITY

SUR-109

ONE MAN EVA 1

ONE MAN EVA I
(MIN TIME)

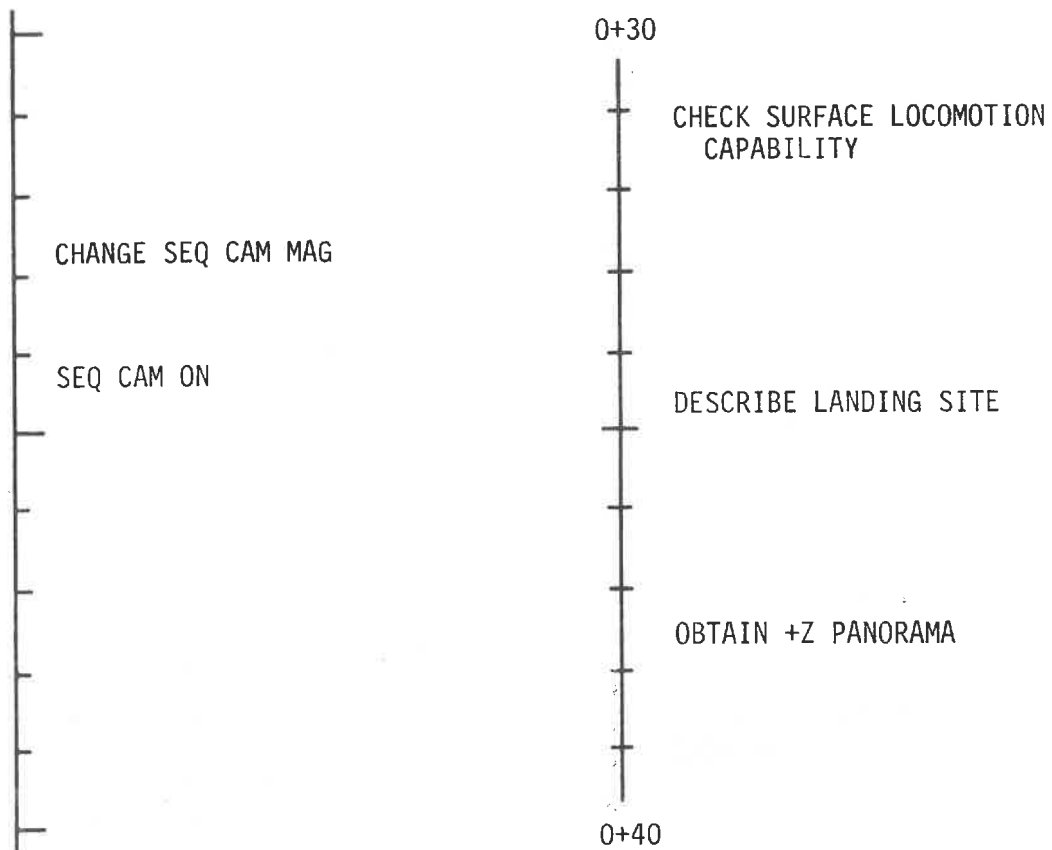


SUR-110

Basic Date 3/23/70

Changed

Basic Date 3/23/70
Changed _____



SUR-111

Basic Date _____

Changed _____

ASSIST CDR	ATTACH 70MM CAMERA AND
CHANGE SEQ CAM MAG	CONTINGENCY SAMPLE TO LEC
SEQ CAM ON	TRANSFER 70MM CAMERA AND
REMOVE 70MM CAMERA AND	CONTINGENCY SAMPLE INTO LM
CONTINGENCY SAMPLE FROM LEC	CLEAN EMU
	PULL LEC FROM LM & DISCARD
	ASCEND LADDER
	INGRESS
	0+50

SUR-112

Basic Date 3/23/70

Changed _____

Basic Date 3/23/70
Changed

LMP ACTIVITIES	EVA TIME	CDR ACTIVITIES
	0+10	MOVE THRU HATCH
PREPARE LEC PASS JETTISON BAG TO CDR		COMM CHECK
		TOSS JETT BAG -Y AREA
PASS LEC TO CDR		DEPLOY LEC
		DESCEND TO LADDER
PHOTO CDR WITH 70MM CAMERA		DEPLOY MESA
		DESCEND TO FOOT PAD
PHOTO CDR WITH 70MM CAMERA		CHECK ASCENT (HOP UP) STEP TO LUNAR SURFACE CHECK AND DISCUSS MOBILITY AND STABILITY
TRANSFER 70MM CAMERA		LEC CAMERA TRANSFER
SEQ CAM ON (FILM CDR WHEN IN FIELD OF VIEW)		UNSTOW CONT. SAMPLE & DEPLOY HANDLE

SUR-113

ONE MAN EVA 1
(FULL TIME)

ONE MAN EVA 1
(FULL TIME)

MSC FORM 2206B (JUL 67)



0+20

COLLECT SAMPLE - STOW
CSRC ON +Z STRUTS
PHOTO CONTINGENCY SAMPLE AREA

REMOVE MESA COVER
UNSTOW & ERECT TV TRIPOD
UNSTOW & MOUNT TV ON TRIPOD
UNSTOW TV CABLE

CARRY TV (50' +Z/+Y) TO
VIEW MESA, S-BAND & FLAG

ERECT MESA TABLE
ATTACH ETB TO MESA TABLE

STOW WEIGH BAGS ON MESA
UNSTOW PLSS BATTS & LiOH
CANS AND PLACE IN ETB

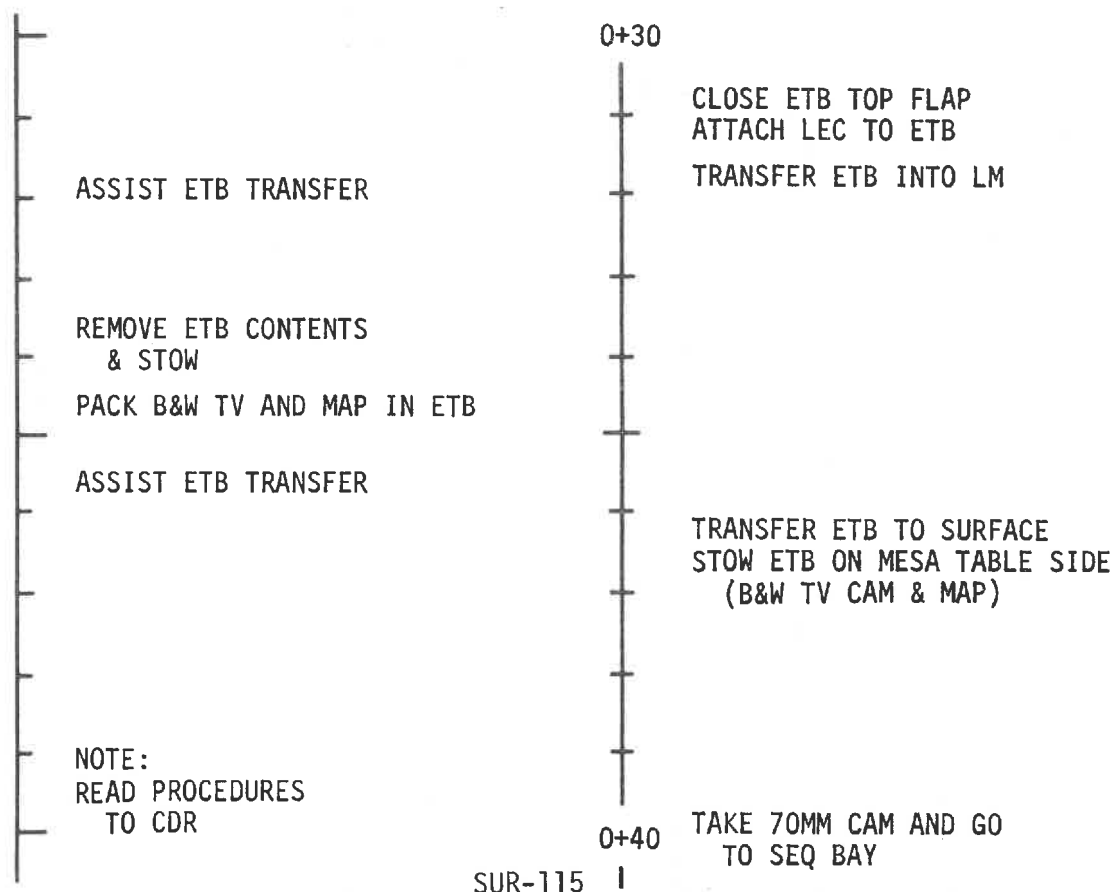
REMOVE SAMPLE FROM CSRC
HANDLE & PLACE IN ETB

0+30

SUR-114

Basic Date 3/23/70
Changed

Basic Date 3/23/70
Changed



Basic Date

Changed _____

CDR ACTIVITIES

0+50

OPEN SEQ BAY DOORS

REMOVE & LOWER PKG #1

DISCONNECT LANYARD & BOOM
STOW LANYARD
REPOSITION PKG #1

STOW BOOM #1

REMOVE & LOWER PKG #2

DISCONNECT LANYARD & BOOM
STOW LANYARD

REPOSITION PKG #2

STOW BOOM #2

REMOVE AND EXPAND ALHTC
RELEASE TOOLS & PLACE IN HTC

SUR-116

CDR ACTIVITIES

1+00

REMOVE DRILL

UHT'S IN PKG'S

MATE MAST AND ATTACH TO
PKG #1

TILT PKG 2 TO GROUND
(FUELING) POSITION

CLOSE SEQ BAY DOORS

DEPLOY CASK TILT LANYARD
TILT CASK AND STOW LANYARD

REMOVE AND DISCARD DOME/DRT

Basic Date 3/23/70

Changed _____

Basic Date 3/23/70
Changed _____

1+10	ENGAGE & CHECK FTT	RETURN TO LM
	WITHDRAW FUEL CAPSULE & FUEL RTG - REPORT TEMP	
	DISENGAGE FTT & DISCARD TILT PKG 2 UPRIGHT & ATTACH TO MAST TO MAKE BAR BELL	REPOSITION TV TO VIEW ALSEP SITE
	CARRY ALSEP TO DEPLOYMENT SITE	GO TO SEQ BAY
	NOTE: REST ENROUTE	CARRY ALHTC & DRILL TO ALSEP SITE >300 FT WEST
	REPORT COMPLETION OF TRAVERSE	
1+10		1+20

SUR-117

Basic Date _____

Changed _____

1+30

SURVEY SITE TO DETERMINE
ALSEP EXPERIMENTS LOCATION

DISCONNECT MAST - PKG 1

RELOCATE PKG 2 10 FT WEST

DISCONNECT MAST - PKG 2

EMPLACE PKG 2 & STOW MAST
REMOVE RTG CABLE REEL
BOYD BOLTS

DEPLOY CABLE - DISCARD REEL

REPORT AMPS AND CONNECT
CABLE

1+40

TILT PKG #1

ALIGN PKG #1 (EYEBALL)

DEPLOY PSE

REMOVE PSE STOOL (PKG 2)
IMPLACE PSE STOOL (10 FT
EAST OF PKG 1 - SCOOP HOLE
WITH UHT AS REQUIRED)

REMOVE SUBPALLET (PKG 2)
RELEASE PSE BOYD BOLTS
REMOVE PSE - PULL OFF
BOLTS - CARRY TO PSE
STOOL

REMOVE GIRDLE - PLACE PSE
ON STOOL
DEPLOY THERMAL SKIRT

SUR-118

Basic Date 3/23/70
Changed _____

Basic Date 3/23/70
Changed _____

1+50	LEVEL PSE	2+00	EMPLACE CPLE (10 FT SOUTH)
	REPORT LEVEL & ALIGNMENT		LEVEL AND ALIGN CPLE (USE UHT AS GNOMON)
	RELEASE CCGE BOYD BOLTS & REMOVE CCGE (PKG 1)		RELEASE HFE BOYD BOLTS
	PULL PIN ON REEL		LIFT HFE FROM C/S
	PLACE CCGE ON LUNAR SURFACE (60 FT SOUTH/WEST CENTRAL STATION) CHECK LEVEL & ALIGN		CARRY HFE TO DEPLOY SITE (30 FT S-E OF C/S)
			DETACH PROBE BOX AND LEAN ON HFE
	RELEASE CPLE BOYD BOLTS & REMOVE CPLE (PKG 1)		CENTRAL STATION: LEVEL & ALIGN
	PULL PIN AND ROTATE CPLE (180°) WHILE HOLDING UHT		DEPLOY ANTENNA CABLE

SUR-119

Basic Date _____

Changed _____

START FRONT CENTER AND RELEASE BOYD BOLTS	INSTALL GIMBAL ON MAST REMOVE HOUSING & DISCARD
RELEASE TWO INNER BOYD BOLTS RELEASE CENTER BOYD BOLT & RAISE SUNSHIELD	INSTALL ANTENNA CHECK C/S ALIGNMENT LEVEL ANTENNA
REMOVE CURTAIN COVERS & DISCARD	ALIGN ANTENNA
CHECK CURTAINS PROPERLY DEPLOYED & VELCROED TO PALLET	ENTER ANTENNA AZIMUTH 16.13 ENTER ANTENNA ELEVATION 6.40
RETRIEVE & INSTALL ANTENNA MAST	WEIGHT PSE SKIRT (IF REQ'D) CHECK ANTENNA LEVEL & ALIGN CHECK EMU
RELEASE AIMING MECHANISM BOYD BOLTS & REMOVE	ACTIVATE CENTRAL STATION REPORT SHORT SW AMPS DEPRESS SHORTING SW CHECK AMPS ZERO
2+10	2+20 PHOTO PAN 15 FT SOUTH OF CENTRAL STATION

SUR-120

Basic Date 3/23/70

Changed _____

Basic Date 3/23/70
Changed _____

TAKE HTC & DRILL TO HFE SITE

ALIGN HFE ELECTRONICS BOX

SPLIT PROBE BOX & LEAN
1/2 BOX ON HF ELECT.

DEPLOY HFE PROBE CABLE
(BOX WITHOUT EMPLACE TOOL)

(ASTRO SW 1 TURN ON AT GND
REQUEST)

LOCATE 1ST BORE HOLE
(2X DIA NEAREST CRATER)

REPOSITION HTC AND DRILL
(IF REQUIRED)

2+30 ASSEMBLE DRILL

ASSEMBLE DRILL
PUSH SWITCH TO TEST

PULL PIN 1
PULL PIN 2

TURN LOCK 3-CCW REMOVE
TURN LOCK 4-CCW-REMOVE
& PULL LANYARD

PUSH RACK LEG FROM CLIP-
EXTEND & LOCK LEGS & BRACE

REMOVE AND INSTALL DRILL
HANDLE TO POWER HEAD
REMOVE RACK-EXTEND 3RD LEG &
PLACE RACK ON SURFACE

PULL PIN 5-SWING COLLAR UP
RESET CHUCK-REMOVE DRILL
REMOVE COVER FROM STEM RACK &
RELEASE VELCRO STRAP

ASSEMBLE ONE STEM SET (WITH
BIT) & INSTALL IN CHUCK

ASSEMBLE 2ND STEM SET-PLACE
IN RACK BIT DOWN

BORE HOLE DRILLING

2+40

SUR-121

Basic Date _____
Changed _____

Changed _____

INSERT DRILL BIT INTO SURFACE

REMOVE THERMAL SHROUD

PUSH INWARD ON BOTH HANDLE
SECTION & DRILL TO HANDLE
HEIGHT OF 28 TO 30 INCHES

RELEASE STEM-1/4 TURN-CCW
-CW BACK, BLIP DOWN, BLIP UP
RESET ADAPTER - INSTALL NEW
STEM TO IMPLANTED STEM - FIT
POWER HEAD ADAPTER TO STEM

REPEAT DRILLING UNTIL 6
STEMS ARE IMPLANTED IN
LUNAR SURFACE

2+50

RETRIEVE HALF BOX (WITH EM-
PLACEMENT TOOL) FROM HFE
ELECTRONICS

DEPLOY HFE PROBE CABLE

LOCATE 2ND BORE HOLE

RETRIEVE DRILL & RACK

DRILL 2ND BORE HOLE
AS PER FIRST BORE HOLE
PROCEDURES

3+00

SUR-122

Basic Date 3/23/70
Changed _____

Basic Date 3/23/70
Changed

3+10 HFE PROBE DEPLOY

SUR-123

REMOVE PROBE & INSERT INTO BORE
HOLE
REMOVE TOOL & EXTEND-CHECKLOCKED
ENGAGE CABLE ABOVE INTERNAL
THERMAL SHIELD
INSERT TOOL AND REPORT MARKS
REMOVE TOOL; PLACE TOP SHIELD
REPORT TOOL MARK FROM GND
TO TOP OF BORE
PROCEED AS ABOVE FOR 2ND PROBE

3+20
(-40)

SHOOT STEREO PR BORE HOLES TO
CEN STA.
RELOCATE DRILL DEBRIS >8 FT SE
RETURN TRAVERSE
(COLLECT SELECTED SAMPLES IF
EVA EXTENDED TO 4+30)

Basic Date

Basic Date _____
Changed _____

REPOSITION TV +Y/50 FT TO
VIEW MESA-GND GIVES SETTINGS

PLACE 70MM CAM IN ETB

ATTACH WEIGH BAG FROM
MESA TO SCALE

SAMPLES FROM HTC INTO
WEIGH BAG

LENS FROM HTC TO ETB

3+30 HTC IN SUN
(-30)

REMOVE SOLAR WIND

CARRY TO DEPLOYMENT SITE

EXTEND STAFF & DEPLOY FOIL

IMPLANT STAFF IN SURFACE

REMOVE CSC - DEPLOY
HANDLE AND PLACE DOWN
SUN (NW/15FT)

OFFLOAD SRC 2 &
PLACE ON +Y FOOTPAD (LID
DOWN SUN)

OFFLOAD SRC 1 & PLACE LID
DOWN SUN

COLLECT SAMPLES FROM
LM AREA & PLACE IN WEIGH BAG

3+40
(-20)

SUR-124

Basic Date 3/23/70
Changed _____

Basic Date 3/23/70
Changed _____

FINISH FILLING WEIGH BAG WITH
LOOSE MATERIAL (COULD LOAD UP
TOTE BAG INSTEAD OF WEIGH
BAG & TRANSFER INTO LM)

CLOSE BAG AND PACK IN ETB
CHECK CONTENTS OF ETB
70MM CAM
LENS
WEIGH BAG(S)

CLOSE ETB FLAP AND HOOK-UP
TO LEC

ETB TRANSFER

3+50
(-10)

SUR-125

Basic Date _____

Changed _____

LMP ACTIVITIES	EVA TIME	CDR ACTIVITIES
STOW ETB CONTENTS	3+50 (-10)	
	+	REMOVE BRUSH FROM MESA
	+	BRUSH LEGS-FRONT OF SUIT
	+	
	+	CLEAN EMU
PASS LEC TO CDR	+	ASCEND TO PLATFORM
ASSIST CDR INGRESS	+	STOW LEC ON PLATFORM (HANG UP ON PLAT-RAIL)
	+	INGRESS

SUR-126

Basic Date 3/23/70

Changed _____

Basic Date 3/23/70
Changed _____

LMP ACTIVITIES	EVA TIME	CDR ACTIVITIES
	0+10	EGRESS
ASSIST CDR EGRESS		MOVE THRU HATCH
PASS EQUIPMENT TO CDR		JETTISON EQUIPMENT
HOOKUP LEC		HAND LEC TO LMP
LOAD ETB		DEPLOY LEC
70MM CAMERA		DESCEND TO LUNAR SURFACE
70MM MAGS		
MAP		COMM CHECK
COMM CHECK		RECONFIGURE TV FOR EVA II
ATTACH ETB TO LEC		
ASSIST ETB TRANSFER		TRANSFER ETB DOWN
PHOTO CDR AS ABLE		STOW ETB ON MESA
		70MM CAM ON RCU
	SUR-127	

CDR ACTIVITIES

0+20 GET SRC 2 FROM FOOTPAD

PLACE SRC 2 ON MESA TABLE
& OPEN

RETRIEVE HTC & PLACE AT MESA

OFF LOAD SRC 2
PLACE IN HTC:
ONE WEIGH BAG
GAS ANALYSIS
SPECIAL ENVIR.
MAGNETIC CONTAINER
35 BAG DISPENSER
CORE TUBES & CAPSCLOSE ORGANIC SAMPLE - IN SRC
STOW TOOLS IN HTC & 70MM MAG
& MAP
TONGS ON YO YO

0+30

CDR ACTIVITIES

PROP UP PHOTOMETRIC CHART
DWN SUNTAKE 4 PHOTOS AT 5 FT NORMAL
TO CHART (f: 5.6,8,11,16)

1 PHOTO f: 11 AT 45° AZIMUTH

RETRIEVE CHART & PLACE IN HTC

CONTAMINATED SAMPLE (QUAD 3)
PLACE GNOMON
PHOTO STEREO PAIR CROSS SUN

PREPARE BAG

COLLECT SAMPLE

PLACE SCOOP ON HTC

DOUBLE BAG SAMPLE &
REPORT NUMBERS & STOW

PLACE GNOMON ON HTC

SUR-128

0+40 TRAVERSE

Basic Date 3/23/70

Changed _____

Basic Date 3/23/70
Changed

CDR CARRIES:
TONGS (ON YO YO)
70MM CAMERA (ON RCU)
HTC:
- TRENCHING TOOL
- GNOMON
- EXTENSION HANDLE
- SMALL SCOOP
- WEIGH BAG
- 3 CORE TUBES & CAPS
- HAMMER
- GAS, MAG & SPEC CONTAINERS
- 35 BAG DISPENSER
- 70MM MAG

NOTE: REPORT -
ALL MOVEMENTS INCLUDING
DIRECTION BETWEEN SAMPLES
LOCATION WRT LM
ALL PHOTOS
SAMPLE BAG NUMBERS

0+50

NOTE:
UNDERLINED ACTIVITIES
OCCUR DURING THE TRAVERSE
BUT NOT NECESSARILY AT
THE TIMES SHOWN

TYPICAL DOCUMENTED
SAMPLE COLLECTION

PLACE ALHTC
GNOMON NEAR SAMPLE
STEREO PAIR (CROSS SUN)
PHOTO SAMPLE (DOWN SUN)
DEPLOY BAG IN DISPENSER
COLLECT SAMPLE, DESCRIBE
& PLACE IN BAG
SEAL BAG AND REPORT NUMBER
STOW SAMPLE

1+00
SUR-129

Basic Date

Basic Date _____
Changed _____

1+10

PHOTO SITE AND HORIZON
15 FT CROSS SUN

PICKUP GNOMON & HTC
AND CONTINUE TRAVERSE

NOTE: TAKE PANORAMAS -
DIRECTION CHANGES > 45°
MAJOR POINTS OF INTEREST
RIMS OF LARGE CRATERS

1+20

1+30

SUR-130

Basic Date 3/23/70
Changed _____

Basic Date 3/23/70
Changed

TYPICAL CORE TUBE
SAMPLE COLLECTION

PLACE HTC

PLACE GNOMON UP SUN

ASSEMBLE CORE TUBE/HANDLE-
REPORT NUMBER

REMOVE HAMMER FROM HTC

DRIVE TUBE INTO SURFACE

PHOTO TUBE & HORIZON CROSS SUN
@ 15 FT

1+40

REMOVE CORE & CAP

REMOVE HANDLE & STOW

STOW SAMPLE IN HTC

PHOTO SITE CROSS SUN
AT 5 FT (STEREO PAIR)

1+50

SUR-131

Basic Date

Changed _____

2+00

POLARIMETRY PHOTOS

LOCATE ROCK CLUMP
& PLACE GNOMON

TAKE 1 PHOTO DWN SUN
15 FT

ATTACH POLARIZING FILTER,
GO TO f5.6, 1/125

MOVE TO CROSS-SUN 5 FT
(90° PHASE ANGLE)

TAKE ONE PHOTO AT EACH
OF 3 FILTER POSITIONS
(L,C,R)

MOVE TO 110° PHASE ANGLE
TAKE ONE PHOTO AT EACH
OF 3 FILTER POSITIONS
(L,C,R)

MOVE TO 130° PHASE ANGLE
TAKE ONE PHOTO AT EACH
OF 3 FILTER POSITIONS
(L,C,R)

GO TO RIM OF CRATER
& TAKE ONE PHOTO AT
EACH OF 3 FILTER POSITIONS
(L,C,R) OF OPPOSITE WALL
OF CRATER (SUNLIT)

DISCARD FILTER & RESET CAMERA
TAKE DOCUMENTED SAMPLES
OF ROCKS FROM CLUMP
(USUAL PHOTO METHOD)

2+10

SUR-132

Basic Date 3/23/70
Changed _____

Basic Date _____

SUR-133

2+20 REMOVE SEAL PROTECTOR &
SEAL SECS & STOW

 $2+30$

Basic Date _____
Changed _____

GAS ANALYSIS SAMPLE
(SMALL CAN)

PLACE GNOMON UP SUN
PHOTO SAMPLE DOWN SUN
PHOTO SAMPLE CROSS SUN
(STEREO PAIR)

RETRIEVE SAMPLE CONTAINER &
OPEN

COLLECT SAMPLE & PLACE
IN GASC

REMOVE SEAL PROTECTOR CLOSE &
SEAL GASC

STOW CONTAINER IN HTC

PHOTO SITE CROSS SUN

LANDMARK PHOTO (IF REQ'D)

2+40

MAGNETIC SAMPLE
(WHITE CAN)

PROCEED AS ABOVE

2+50

SUR-134

Basic Date 3/23/70
Changed _____

Basic Date 3/23/70
Changed

NOTE: COLLECT DOCUMENTED
SAMPLES AS APPLICABLE
DURING RETURN TO LM

NOTE: LMP
READ PROCEDURES
TO CDR & PHOTO

3+00

REPORT TRAVERSE COMPLETE

EVA CLOSEOUT

PLACE IN SRC 2:
CORE TUBES
MAG SAMPLE
GAS SAMPLE
SPEC SAMPLE
DOCUMENTED SAMPLES
FINES & ROCKS FROM
AROUND THE LM

3+10 CLOSE & SEAL SRC 2 &
PLACE ON LM FOOTPAD

SUR-135

Basic Date

Changed _____

3+10 CLOSE & SEAL SRC 2 &
PLACE ON LM FOOTPAD

FILL SRC 1 WITH FINES &
ROCKS FROM THE LM AREA

NOTE: IF SRC 1 NOT USED IN 1ST
EVA

RETRIEVE SRC 1 & PLACE ON
MESA TABLE

OPEN SRC 1 & PLACE SCALE
ON MESA-WEIGH BAG ON SCALE

DISCARD CORE STEMS AND CAPS

3+20

3+30

SUR-136

Basic Date 3/23/70

Changed _____

Basic Date 3/23/70
Changed

DEPLOY CLOSE UP CAMERA &
TAKE PICTURES AROUND THE LM

OFF LOAD CLOSE CAMERA FILM
AND STOW CASSETTE IN ETB

RETRIEVE SOLAR WIND - BAG &
STOW IN ETB

3+40

SUR-137

Basic Date _____
Changed _____

LMP ACTIVITIES	EVA TIME	CDR ACTIVITIES
		STOW 70MM CAM & MAGS IN ETB
		REMOVE SEAL PROTECTOR-CLOSE & SEAL SRC 1
		CLOSE ETB FLAP
ASSIST ETB TRANSFER		TRANSFER ETB INTO LM
STOW ETB		TRANSFER HOOKS TO SURFACE
ASSIST SRC 1 TRANSFER		TRANSFER SRC 1 INTO LM
STOW SRC 1		TRANSFER HOOKS TO SURFACE
ASSIST SRC 2 TRANSFER	3+50 SUR-138	TRANSFER SRC 2 INTO LM

Basic Date 3/23/70
Changed _____

Basic Date 3/23/70
Changed

— ASSIST SRC 2 TRANSFER
— DISCONNECT LEC
— TEMP STOW SRC 2
—
— HAND LEC TO CDR
—
— ASSIST CDR INGRESS
—
—

3+50 TRANSFER SRC 2 INTO LM
+ DISCARD TONG
+ CLEAN EMU
+ ASCEND TO PLATFORM
+ JETTISON LEC
+ MOVE THRU HATCH
+
+

SUR-139

Basic Date

Basic Date 3/23/70
Changed

POST ONE-MAN EVA

Perform POST EVA 1 or 2 As Applicable

SUR-140

ONE MAN EVA POST

Basic Date 3/23/70
 Changed _____

EMERGENCY LAUNCH STOWAGE
BEFORE EVA 1

PLSS,OPS,RCU (BOTH)	DOFF
CDR PLSS	RECHARGE STA (REMOVE ADJ STRAP)
OPS (BOTH)	FLOOR (TEMPORARY)
JETT BAG	ISA
RCU (BOTH)	ISA
70 MM CAMR & BRKT (BOTH)	ISA
LEC	ISA
SUR SEQ CAMR	ISA
TV CAMR	ISA
EV GLOVES	HELMET BAG
LEVA	HELMET BAG
LM EVA ANTENNA - STOW	
UTILITY LIGHTS (BOOT BOX)	INSTALL
ISA	HUNG AFT
LMP PLSS.	DONNING STA (USE 2 ADJ STRAPS ON
	BOTTOM. USE WAIST TETHERS TO PREVENT
	LATERAL MOVEMENT)
OPS (BOTH)	FLOOR (FOR LAUNCH)
HELMET BAG	FLOOR
FWD HATCH - CLOSE & LOCK	

SUR-141

EMER LAUNCH

EMER LAUNCH
STOWAGEEMERGENCY LAUNCH STOWAGE
BEFORE EVA 2

PLSS, OPS, RCU (BOTH)	DOFF
CDR PLSS	RECHARGE STA (REMOVE ADJ STRAP)
OPS (BOTH)	FLOOR (TEMPORARY)
EV GLOVES, LEVA	HELMET BAG
SRC	SRC AREA
LM EVA ANTENNA - STOW	
UTILITY LIGHTS (BOOT BOX)	INSTALL
RCU (BOTH)	UPPER BOOT BOX
SUR SEQ CAMR	LOWER BOOT BOX
PRI LiOH CART & BRKT	ENG COVER OR JETT BAG
PLSS BATT & CART	LHSSC
FEEDWATER BAGS	JETT BAG
70MM CMR, HANDLE, TRIG (BOTH)	RIGHT SIDE (ONE MAG IN FECAL EMESIS AREA)
RCU/CMR BRKT	JETT BAG
RETURN ITEMS	ISA
ETB & CONTENTS	ISA
ISA	HUNG AFT (FROM RCU SHELF TO ENG COVR)
LHSSC	HUNG AFT (FROM SLEEP STA FITTINGS TO ENG COVR)
TOTE BAG	TIE TO FRONT OF LHSSC WITH TOTE BAG STRAPS
HAMMOCKS	LEFT SIDE OR JETT BAG
LMP PLSS	DONNING STA (USE 2 ADJ STRAPS ON BOTTOM. USE 2 WAIST TETHERS TO PREVENT LATERAL MOVEMENT)
OPS (BOTH)	FLOOR (SECURE)
JETT BAG	TIE TO FLOOR (PUT BOOTS IN BAG TO PROTECT Z-27 BULKHEAD)
HELMET BAGS (BOTH)	FLOOR
FWD HATCH - CLOSE & LOCK	SUR-142

Basic Date 3/23/70Changed 3/31/70

NASA — MSC

Basic Date 3/23/70
Changed _____

EMERGENCY LAUNCH STOWAGE
BEFORE EQUIPMENT JETTISON

PLSS,OPS,RCU (BOTH)	DOFF
CDR PLSS	RECHARGE STA (REMOVE ADJ STRAP)
OPS (BOTH)	FLOOR (TEMPORARY)
EV GLOVES, LEVA	HELMET BAG
SRC	SRC AREA
UTILITY LIGHTS (BOOT BOX)	INSTALL
LM EVA ANTENNA - STOW	
RCU (BOTH)	LHSSC
70 MM CAMR, HANDL, TRIG	RIGHT SIDE(ONE MAG IN FECAL EMESIS AREA)
RCU/CAMR BRKT	JETT BAG
RETURN ITEMS	ISA
ETB & CONTENTS	ISA
TOTE BAGS	ISA (TIE SECOND BAG TO ISA IF REQ'D)
ISA	HUNG AFT
LHSSC	JETT BAG
LMP PLSS	DONNING STA (USE 2 ADJ STRAPS ON BOTTOM. USE 2 WAIST TETHERS TO PREVENT LATERAL MOVEMENT)
OPS (BOTH)	FLOOR (FOR LAUNCH)
JETT BAG	TIE TO FLOOR (PUT BOOTS IN BAG TO PRO- TECT Z-27 BULKHEAD)
HELMET BAG (BOTH)	FLOOR
FWD HATCH - CLOSE & LOCK	

SUR-143

Basic Date _____
Changed _____

LM-7

Basic Date 3/23/70
Changed

EMERGENCY LIFT-OFF

IF TIME PERMITS:
Don Helmets And Gloves
PRESS REG A&B - EGRESS
SUIT GAS DIV - PULL/EGRESS
SUIT CIRCUIT RELIEF - AUTO

PGNS ACT

PRO

(RESTART Lt-On, STBY Lt-Off)

RSET

V96E

CB(11) IMU OPR - Close

(NO ATT Lt-Off In 90 Sec)

Verify:

CB(16) INV 2 - Close

INV-2

Bat 5&6 - ON tb-gray

BAT 1&3 - OFF/RESET tb-bp

DES H20 - CLOSE

ASC H20 - OPEN

WATER TANK SEL - ASC

CABIN REPRESS - CLOSE

DES 02 - CLOSE

ASC 02 No. 1 - OPEN

Configure CBs Per EMER LIFT OFF Status Charts

SUR-144

EMER LIFT OFF

The diagram illustrates the aircraft's electrical system, organized into several main sections:

- AC BUS B:** Includes components like SE WIND, H₂ POOL, PROP, DEP, S-DO, AWT, OMBREL, ADR, AOT LAMP, SE PDAI, HUAN, LTO, BWS 2, BWS 1, BWS 3, BWS 1, BWS 2, BWS 1, AC BUS VOLT, CDR WIND, HTR, TAPL, SCOR, AOT LAMP, BWS 2, DECA, CDR, BWS 1, LTO.
- AC BUS A:** Includes components like SE WIND, H₂ POOL, PROP, DEP, S-DO, AWT, OMBREL, ADR, AOT LAMP, SE PDAI, HUAN, LTO, BWS 2, BWS 1, BWS 3, BWS 1, BWS 2, BWS 1, AC BUS VOLT, CDR WIND, HTR, TAPL, SCOR, AOT LAMP, BWS 2, DECA, CDR, BWS 1, LTO.
- RCS SYS A:** Includes components like RAMP, SOV, QUAD 4, TCA, QUAD 3, TCA, QUAD 2, TCA, QUAD 1, TCA, ASC, PBD 2, ASC, PBD 1, THRUST, AMBUSH, TURRET, CDR, S-4412, BWS 2, BWS 1, BWS 3, BWS 1, BWS 2, BWS 1, AC BUS VOLT, CDR WIND, HTR, TAPL, SCOR, AOT LAMP, BWS 2, DECA, CDR, BWS 1, LTO.
- FLIGHT DISPLAYS:** Includes components like BWS 2, BWS 1, BWS 3, BWS 1, BWS 2, BWS 1, AC BUS VOLT, CDR WIND, HTR, TAPL, SCOR, AOT LAMP, BWS 2, DECA, CDR, BWS 1, LTO.
- AC BUS A:** Includes components like SE WIND, H₂ POOL, PROP, DEP, S-DO, AWT, OMBREL, ADR, AOT LAMP, SE PDAI, HUAN, LTO, BWS 2, BWS 1, BWS 3, BWS 1, BWS 2, BWS 1, AC BUS VOLT, CDR WIND, HTR, TAPL, SCOR, AOT LAMP, BWS 2, DECA, CDR, BWS 1, LTO.
- HEATERS:** Includes components like PROP, BWS 2, BWS 1, BWS 3, BWS 1, BWS 2, BWS 1, AC BUS VOLT, CDR WIND, HTR, TAPL, SCOR, AOT LAMP, BWS 2, DECA, CDR, BWS 1, LTO.
- INST:** Includes components like BWS 2, BWS 1, BWS 3, BWS 1, BWS 2, BWS 1, AC BUS VOLT, CDR WIND, HTR, TAPL, SCOR, AOT LAMP, BWS 2, DECA, CDR, BWS 1, LTO.
- STAB/CONT:** Includes components like BWS 2, BWS 1, BWS 3, BWS 1, BWS 2, BWS 1, AC BUS VOLT, CDR WIND, HTR, TAPL, SCOR, AOT LAMP, BWS 2, DECA, CDR, BWS 1, LTO.
- ED:** Includes components like BWS 2, BWS 1, BWS 3, BWS 1, BWS 2, BWS 1, AC BUS VOLT, CDR WIND, HTR, TAPL, SCOR, AOT LAMP, BWS 2, DECA, CDR, BWS 1, LTO.
- LTO:** Includes components like BWS 2, BWS 1, BWS 3, BWS 1, BWS 2, BWS 1, AC BUS VOLT, CDR WIND, HTR, TAPL, SCOR, AOT LAMP, BWS 2, DECA, CDR, BWS 1, LTO.
- HEATERS RCS SYS A/B-1:** Includes components like QUAD 4, QUAD 3, QUAD 2, QUAD 1.
- ECS:** Includes components like SUT, FAN 1, CABIN, FAN 1, OILCOOL, PUMP, 2, AUTO, TEMPS, 1.
- COMM:** Includes components like UP, DATA, LINES, SDC, S-4412, PWR, AMP, 1, VWP, B, TETS, VWP, A, SCYE, CDR, ALARM, SIO, SPS, BWP, LPO, SDR, BWS, SDR, LBC, SDR, SIO, SPS.
- BPS:** Includes components like BWT, PBD 2, BWS, 1, CROSS, TIE, BUS, 1, LUNAR, BUS, TIE, DES, RCA, CONT, DES, RCA, DES, RCA, CONT, BWS, 1, PC, BUS, VOLT.

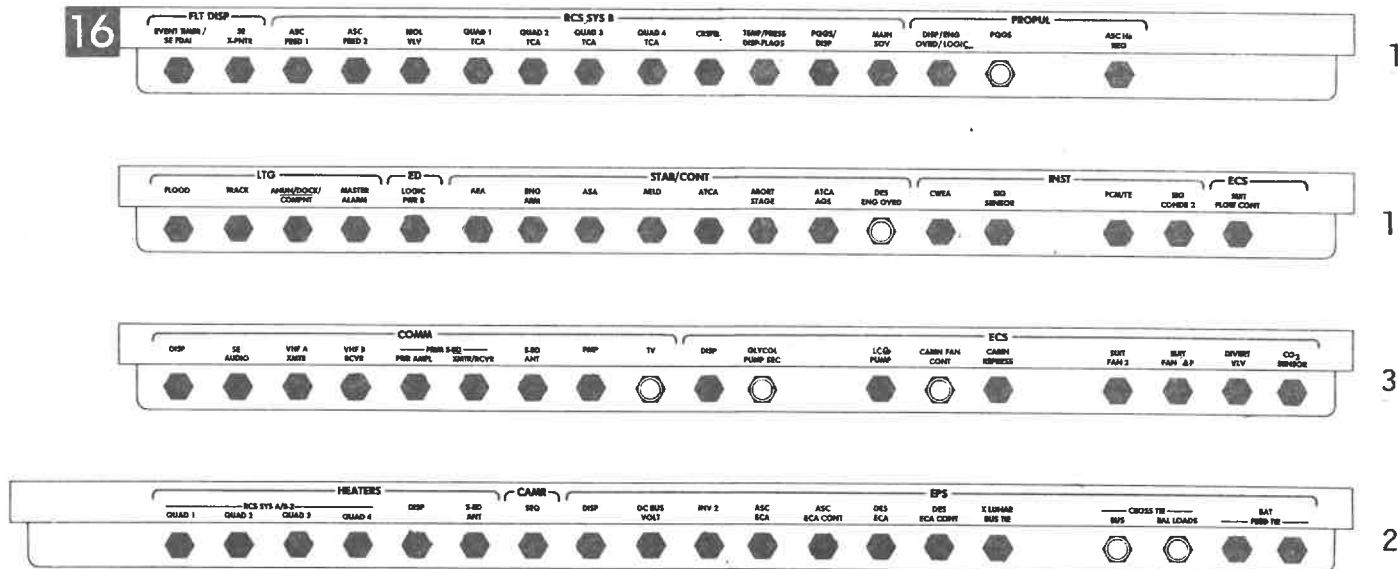
LM-7

Basic Date 3/23/70
Changed

LM-7

Basic Date 3/23/70
 Changed _____

EMER LIFT OFF



SUR-146

Basic Date _____
 Changed _____

Changed _____

MSFN-UPDATE
UPDATA LINK - DATA
(MSFN Updates
State Vector)
UPDATA LINK - VOICE BU

AGS ACTIVATION

AGS STATUS - OPERATE (Master
Alarm, AGS Warn Lt-On)

ALIGN PGNS
P57E, OPT 4, PRO

N34 0, 0, 0 Present Time, PRO

N06 R1 00010
00001
00010

PRO
(NO ATT Lt - On/Off, Twice)
N04 + _____ Δ TILT
PRO

N22 ICDU ANGLES
PRO (NO ATT LT - On/Off)

N05 _____ Angle Diff
PRO

N93 Torquing Angles _____ X
PRO _____ Y
SUR-121 _____ Z SUR-147

LM-7

Basic Date _____ 3/23/70
Changed _____

LM-7

Basic Date 3/23/70
Changed _____

N25 00014 ENTR
P00E

AGS INITIALIZATION

V16 N65E, 377 _____

V47E, 414+1E

400	+4E	Lunar Align
240	+56978	X Position Comp
261	<u>-00013</u>	Y Velocity Comp
262	<u>-00151</u>	E Z Velocity Comp
254	_____	E Epoch Time(377R)
414	+ 2E	Nav. Initial Via DEDA

400 + 3E
413 + 1E

P _____, Y _____ (+110/-41)
S-BAND ANT-SLEW (>3.0)
TRACK MODE - AUTO (>4.0)

RCS HOT FIRE
CDR ACA (Deflect Out-of-Detent)
Roll Rt, Lt
Pitch Up, Dn
Yaw Rt, Lt

SUR-148

Basic Date _____

Changed _____

TARGET PGNS

GUID CONT - PGNS

MODE CONT (Both) - AUTO

P12E

N33 _____:_____ TIG ASC

PRO

N76 _____ VH FINAL (+55095)
_____ HDOT FINAL (+00195)
_____ XRNG (+00000)

PRO

N74 _____:_____ TFI
_____ YAW
_____ PITCH

TARGET AGS

224 + _____ (58217)

226 + _____ (58217)

232 + _____ (+00600)

410 + 0

411 + 1

465 + _____ (+00195)

514 _____ (-60000)

515 _____ (-44223)

516 _____ (+00000)

662 + 0E

673 + 0E

SUR-149

LM-7

Basic Date _____ 3/23/70

Changed _____

LM-7

Basic Date 3/23/70
Changed _____

MASTER ARM - ON
ASC He SEL - BOTH
ASC He PRESS - FIRE
MASTER ARM - OFF
SYS A&B ASC FEED 2 (2) - OPEN
SYS A&B MAIN SOV (2) - CLOSE
CRSFD - OPEN

ENABLE CONTROLS

ACA PROP (Both) - ENABLE
ACA/4 JET (Both) - ENABLE
ATT CONT (3) - MODE CONT
MODE CONT (Both) - AUTO
TTCA/TRANSL (Both) - ENABLE
MODE SEL - AGS
RNG/ALT MON - ALT/ALT RT

CONFIGURE COMM

VHF A: XMTR - VOICE/RNG
 : RCVR - ON
VHF B: RCVR - ON
AUDIO (Both): VHF A - T/R
 VHF B - RCV

BEGIN FINAL COUNTDOWN

T-5:00 BATS 2&4 - OFF/RESET tb-bp
 DES BATS - DEADFACE tb-bp
 Check APS Start Card

GO TO LM TIMELINE BOOK

SUR-150

Basic Date _____
Changed _____

LM-7

Basic Date 3/23/70
Changed

MANUAL ASCENT

PRO
(RESTART Lt-On, STBY Lt-Off)
RSET
V96E

CB(11) IMU OPR - close
(NO ATT Lt-Off in 90 sec)

If Time Allows:
V41N20E
Load ICDU Angles That
Existed After Touchdown
V40 N20E (Releases IMU)

BAT 5,6 - ON, tb-gray
BAT 1,2,3,4 - OFF/RESET, tb-bp

CB(11) AGS - Close
AGS STATUS - OPERATE
(Master Alarm, AGS Warn Lt-On)

400 + 4E Lunar Align
Wait 30 sec Before:
400 + 0E

MODE CONTROL (Both) - ATT HOLD

Configure CB's Per EMER LIFT OFF
Status Charts (Pages SUR-145 & 146)

P , Y (+110/-41)
S-BAND - SLEW (>3.0)
TRACK MODE - AUTO (>4.0)

SUR-151

MANUAL ASCENT

MANUAL ASCENT

CDR ACA (Deflect Out of Detent)

Roll-Rt, Lt

Pitch-Up, Dn

YAW-Rt, Lt

MASTER ARM - ON

ASC He SEL-BOTH

ASC He PRESS - FIRE

Go To APS START CARD

SUR-152

LM-7

Basic Date 3/23/70

Changed

NASA — MSC